

REVIEW ARTICLE | APRIL 01 2025

Shining light on devices: New perspectives in non-volatile memory device and material investigation

Atif Jan  ; Dawn M. Kelly  ; Giuliana Di Martino  



APL Electronic Devices 1, 021501 (2025)

<https://doi.org/10.1063/5.0255061>



Articles You May Be Interested In

CBRAM based on CVD-grown MoSe₂ with the coexistence of volatile and non-volatile resistive switching

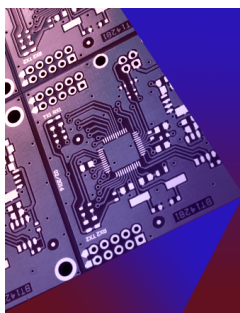
J. Appl. Phys. (July 2025)

Non-volatile materials for programmable photonics

APL Mater. (October 2023)

Volatile threshold switching devices for hardware security primitives: Exploiting intrinsic variability as an entropy source

Appl. Phys. Rev. (May 2024)



APL Electronic Devices

Fostering connections across multiple disciplines
in the broad electronics community

Now Open for Submissions



Shining light on devices: New perspectives in non-volatile memory device and material investigation

Cite as: APL Electron. Devices 1, 021501 (2025); doi: 10.1063/5.0255061

Submitted: 26 December 2024 • Accepted: 27 February 2025 •

Published Online: 1 April 2025



Atif Jan,  Dawn M. Kelly,  and Giuliana Di Martino^{a)} 

AFFILIATIONS

Department of Materials Science and Metallurgy, University of Cambridge, 27 Charles Babbage Rd., Cambridge CB3 0FS, United Kingdom

^{a)} Author to whom correspondence should be addressed: gd392@cam.ac.uk

ABSTRACT

Non-volatile memory (NVM) technologies, including resistive random access memory (RAM), 2D memristors, and ferroelectric RAM, offer major improvements in data storage and computational systems. Despite tremendous promise, these technologies present several obstacles to large-scale commercialization, especially related to material stability, switching processes, and performance consistency. Conventional methods such as transmission electron microscopy, scanning electron microscopy, scanning tunneling microscopy, conductive atomic force microscopy, and XPS, among several others, provide basic insights into structural and compositional features. However, these approaches typically demonstrate limits in recording dynamic, ambient measurements, in-operando opto-electrical, and quantum-scale processes. Emerging characterization methods, such as nanoscale plasmonic microscopy, plasmon-enhanced probe microscopy, interferometric scattering microscopy, and atomic plasmonic switches, provide atomistic, ambient temperature and pressure, simultaneous optical and electrical probing methods with competitive resolution and sensitivity. These approaches allow real-time monitoring of nanoscale electrical transitions, defect dynamics, and interfacial morphology, which are important to understanding NVM device performance. This review presents current and innovative characterization methodologies for comprehensively assessing NVM material qualities and operational features and suggests that enhanced characterization approaches are critical for unraveling underlying processes influencing NVM functioning. By combining several analytical approaches, researchers may solve important difficulties and speed the development of next-generation non-volatile memory systems.

© 2025 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.1063/5.0255061>

I. INTRODUCTION

Over the past 65 years, CMOS devices owing to increased demand for computing and miniaturization have pushed Moore's law to its limit.¹ A significant amount of scientific effort has since been aimed toward alternative technologies with competitive or lower power consumption, higher speed, and enhanced endurance, with the International Roadmap for Devices and Systems (IRDS) suggesting that from 2025, reduced power consumption and heterogeneous integration will be the main drivers in the industry.² Novel developments include neuromorphic architectures inspired by brain structures, combining logic and memory,

which have particular relevance given the recent surge of interest in new approaches to computing for artificial intelligence (AI). To address this momentous challenge, alternatives to CMOS, like oxide-based metal-insulator-metal (MIM) and heterojunction devices, are at the forefront.³ With the advent of oxide electronics, (Ba, Sr)TiO₃-based capacitors,⁴ lead zirconate titanate-based transducers,⁵ and hafnium (Hf)-based oxinitrides for high-k gate insulators⁶ mark significant landmarks in the development of functional oxide materials, particularly aimed at low-voltage applications and high scalability.

Materials that can be reversibly and reliably switched between electrical resistance states can be employed as non-volatile

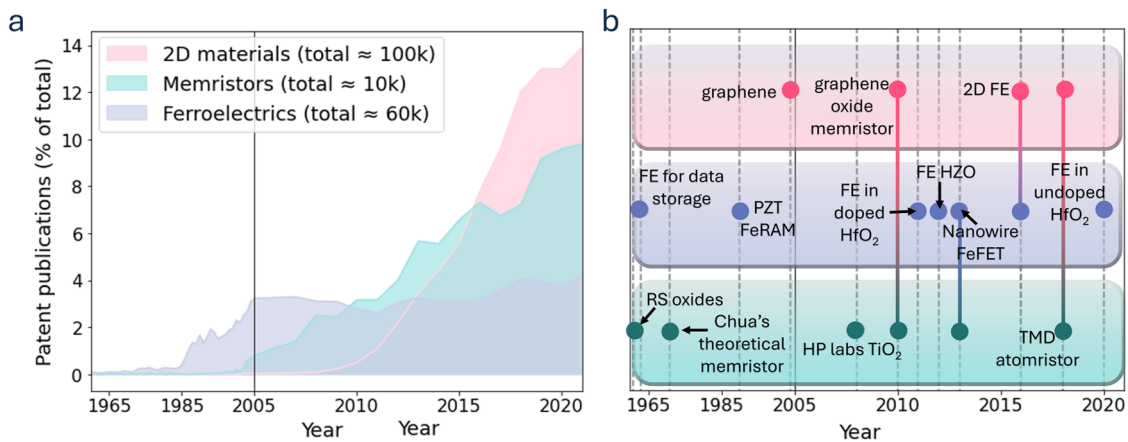


FIG. 1. (a) Number of patents published per year as a percentage of the total to date, using keywords related to two-dimensional materials, memristors, two-dimensional materials, and ferroelectrics. The time axis is compressed before 2005. (b) Key milestones for memristors, two-dimensional materials, and ferroelectrics are highlighted, using the same time axis. Vertical lines mark milestones that fall under multiple categories.

memory (NVMe) devices to encode information.⁷ In 1971, Leon Chua proposed a resistor with memory, or a “memristor,” as a theoretical fourth fundamental circuit element.^{8,9} Although this original theoretical definition involves magnetic induction,¹⁰ the term “memristor” is commonly applied to any resistive switching memory.¹¹ Memristors are generally formed by a two-terminal device in an MIM structure. Under bias, they may be switched between a low resistance state (LRS) and a high resistance state (HRS).

While such resistive switching behavior was first reported in oxides in 1962,¹² the demonstration of a memristor based on TiO₂ in 2008 by Hewlett-Packard Labs sparked interest in resistive random access memory (RRAM) and the possible use of memristors to simulate synaptic functions,¹³ resulting in a significant surge in number of patents published for technologies referring to memristive function in the early 2000s [Fig. 1(a)]. Non-volatile memories (NVMe) are emerging as a feasible alternative to bridge the performance gap between traditional memory technologies like DRAM and flash. NVMe includes memory classes like resistive, ferroelectric (FE), and 2D materials, each offering unique advantages.

The second class of NVMe popular today is based on ferroelectrics. Ferroelectricity, the inherent electric polarization of certain minerals, was first identified in Rochelle salt by Valasek in 1920.¹⁴ Memory devices based on ferroelectric (FE) materials were first proposed in 1952.¹⁵ Ferroelectric materials exhibit a switchable electric polarization, which can be utilized as a parameter to save information.¹⁶ In 1963,¹⁷ a novel solid-state memory resistor was introduced, designed for variable gain components, leveraging ferroelectric material memory capabilities. It utilized remanent polarization to control semiconductor field-effect conductance. The ability of ferroelectric materials to retain a polarization state even in the absence of an externally applied field¹⁸ means they can be used for non-volatile static memory functionality, which was studied throughout the 1970s–1980s using materials including ABO₃ perovskites such as BaTiO₃ (BTO) and lead zirconate-titanate Pb(Zr,

Ti)O₃ (PZT).^{19–21} A series of milestone developments in ferroelectric technologies in the early 2010s [Fig. 1(b)], including the discovery of ferroelectricity in doped HfO₂ in 2011,²² resulted in increased interest by the NVMe community [Fig. 1(a)]. While Pb(Zr_{1-x}Ti_x)O₃ and BaTiO₃, as mentioned, are limited in scaling down film thickness,^{23,24} the HfO₂-based devices show excellent ferroelectric properties at sub-10 nm thicknesses.^{25–31} Moreover, HfO₂ offers a distinct advantage over PZT due to its lead-free composition, mitigating environmental concerns associated with the toxicity of lead-containing materials and compatibility with CMOS technology processing.⁵

Third, the discovery of two-dimensional (2D) or Van der Waals (VdW) materials in the early 2000s is of great significance for achieving high-density integration of devices. VdW materials have a layered structure with covalent in-plane bonding but weaker Van der Waals-type interplanar bonds. This strong in-plane bonding gives mechanical strength while maintaining flexibility, and the weak interplanar bonds mean that layers can be cleaved and separated. Monolayer graphite, now commonly referred to as graphene, was studied both theoretically and experimentally. Early theoretical studies date back to the 1940s³² and experimental observations, including via transmission electron microscopy (TEM), were reported decades prior to the mechanical exfoliation technique introduced by Novoselov *et al.*³³ The implementation of 2D materials into electronics was rapidly investigated following the discovery of graphene in 2004,³³ as is evident in Fig. 1(a), with multiple memristors utilizing graphene oxide as the active material demonstrated by 2010.^{34–37} Closely following this, various electronic devices were constructed incorporating single or few layers of semiconductive MoS₂, a VdW transition metal dichalcogenide (TMD).^{38–42} VdW interactions between 2D materials like graphene, hexagonal boron nitride (hBN), black phosphorus (BP), and 2D TMDs do not require lattice matching, making heterostructures easy to form and regulate switching characteristics via interface control.⁴³ VdW heterostructures have been used to fabricate devices with optical transparency

and ultralow switching powers,^{44–47} while materials including HfSe_2 and MoS_2 have been successfully integrated into crossbar arrays.^{48–50} In addition, with demonstrations of FE in SnTe ⁵¹ and CuInP_2S_6 ,⁵² Among others,^{52–59} 2D FE VdW-based NVMe offers novel functionalities such as multi-level storage, synaptic-like behavior, and neuromorphic computing applications.

This review explores emerging non-volatile memories as functional devices and the concomitant challenges that are now being investigated in recent literature. The obstacles, including unpredictability in operating parameters, insufficient comprehension of switching processes, the influence of vacancies and other defects, inadequate interface management between the active layer and functional electrode, and many intrinsic physical limits, impede the progress of non-volatile memories. This study examines both established and novel characterization approaches that may tackle these problems and promote the advancement of NVMe technology. Section II provides an overview of the basic NVM devices elaborating on their baselines working mechanisms and the major challenges impeding large scale commercialization back. Section III illustrates how these challenges are investigated and addressed using established techniques like electron, probe and optical microscopy techniques. Section IV introduces the novel approaches for atomistic, operando measurements in ambient environment, offering insights into the challenges and potential solutions toward commercialization of non-volatile memory systems.

II. NONVOLATILE MEMORY (NVM) DEVICES: WONDERS AND CHALLENGES

Oxide-based NVMeS are very promising materials in the emerging NVMe applications, yet, many obstacles hinder large-scale commercialization.^{6,25,30} Addressing these roadblocks requires insight into fundamental device dynamics. For commercial relevance, a high degree of consistency from device to device is required, as well as reliable switching over time.⁶⁰ Integration and miniaturization are also key challenges that must be addressed for widespread implementation. Technical challenges include scaling down active layers while maintaining correct functionality, non-uniformity, poorly understood switching mechanisms, 3D monolithic integration, endurance, and reliability.⁶¹

A. Resistive NVMeS

Resistive switching memories, a desirable non-volatile memory technology, depend on the regulated formation and dissolution of conductive filaments (CFs) inside a dielectric layer. The creation and elimination of an atomic conductive filament (CF) inside a device underlies the switching process of resistive memory. In a $\text{Ni}/\text{HfO}_2/\text{SiO}_x/\text{Si}$ memristor, Fig. 2(a) schematically illustrates the evolution of the metal nanofilament in RRAM devices across different operational stages. It depicts the transition from the SET state to the RESET state. The current flows from the Ni electrode to the n^+ -doped Si substrate. The resistive switching mechanism is governed by ion migration within the dielectric layers, leading to the formation and rupture of conductive filaments. During the SET process, voltage application induces the

formation and migration of oxygen vacancies, establishing a conductive path. Simultaneously, Ni atoms from the top electrode diffuse downward, forming an MNF with a truncated cone morphology spanning the SiO_x and HfO_2 layers. In the RESET stage, a high current through the MNF generates Joule heating, promoting thermally assisted Ni diffusion and localized melting at narrow regions of the filament. This leads to an abrupt filament rupture, switching the device to its OFF state. After RESET, residual Ni remains in the silicon as a compound, while disconnected Ni fragments persist in the dielectric layers, potentially serving as nucleation sites for subsequent metal nanofilament formation in the next SET cycle.⁶²

Parameters used to describe the quality of resistive switching devices include switching speed, switching energy cost, on/off ratios, retention, and endurance. Conventional memristors generally use bulk oxides as their active layer. Table I summarizes the performance metrics for some exemplar memristive devices using oxides. A detailed comparison of Table I clearly reveals significant variations in performance metrics when using the same material with different electrodes, as well as when using different materials with the same electrodes.

Resistive memories, although a promising candidate for in-memory computing and energy-efficient neuromorphic devices to meet the computational requirements of future applications, are held back by high variability: poor cycle-to-cycle and device-to-device uniformities, eventually limiting their mass production.⁶¹ The difficulties of reliability and variability in RRAM (evident in Table I) arise from the stochastic formation of conductive filaments, which might differ in shape and position, thereby affecting the device's resistance state. Variability in device performance especially in endurance and memory windows among the most researched materials— TiO_x ,^{80,81} NiO_x ,⁸² ZnO_x ,^{83,84} HfO_x ,^{25,81,85} WO_x ,^{86,87} and ZrO_x ^{88,89}—produces a bigger challenge in understanding these multi-functional materials. Parameter variability in set and reset values spread over a voltage range [Fig. 2(b)] causes stochastic switching dynamics, which leads to erratic device performance in resistive memory devices. Desired values for ideal memory of $>10^{12}$ cycles are needed (although limited, some studies have reported $\text{Pt}/\text{Ta}_2\text{O}_5\text{-x}/\text{Pt}$ memristors exhibiting endurance on the order of 10^{12} cycles⁹⁰) to produce a technologically relevant product.⁹¹ Here, endurance failure often results from the formation of high-resistance interfacial layers or the enlargement of ion migration zones, which may further impair the switching mechanism and intensify device inconsistency.⁹² In addition, the endurance-retention dilemma presents a prevailing challenge. The current-retention conundrum is a significant factor for low-power applications.⁹³ A broad filament can attain substantial retention at the expense of power, while a thin filament can accomplish minimal power usage at the expense of retention. The current retention difficulty is a significant issue that requires deeper examination. *In situ* examination of filamentary and interfacial switching processes is still being investigated. However, owing to the smaller dimensions of the region of interest in filamentary processes, the investigation is hampered by physical and phenomenological challenges.

The optimization of the device metrics discussed above requires knowledge of the fundamental switching mechanism, a detailed investigation of device evolution during cycling, and a

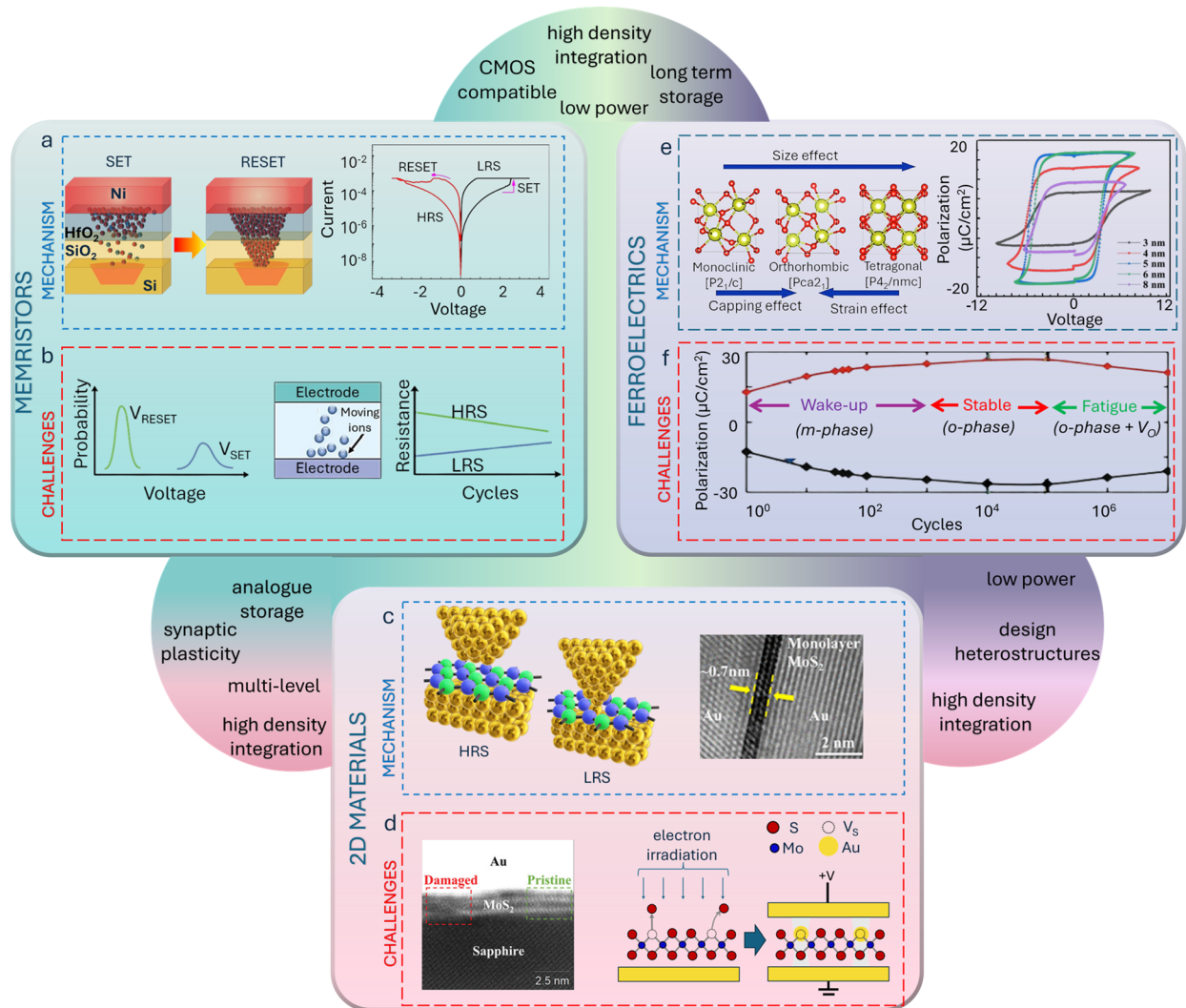


FIG. 2. Resistive memories: (a) Schematic representation of the formation and disintegration of Cu-atomic CF [Reproduced with permission from Wu *et al.*, Adv. Electron. Mater. **1**, 1500130 (2015). Copyright 2015, Wiley-VCH] and I-V characteristics of the TiO_x-based memristor device exhibiting a bipolar switching mode [Reproduced with permission from Kwon *et al.*, Nano-Micro Lett. **14**, 58 (2022). Copyright 2022, Springer⁶³]. (b) Variability in the parameters of set (HRS) and reset (LRS) voltages (left), and characteristic endurance failure behaviors in memristors (right). Potential microscopic causes of cycle endurance failure include the variability of conduction filaments (middle) and the movement of mobile ions (blue spheres), which results in cycle endurance failure. 2D memristors [Reproduced with permission from Sun *et al.*, Nat Commun **10**, 3453 (2019). Copyright 2019, Springer]. (c) Illustration of "atomistor" style CF across a monolayer of hBN for vertical two-terminal memristor, and a TEM cross-section of a memristor based on a MoS₂ monolayer showcasing atomically flat electrode interfaces [Reproduced with permission from Ge *et al.*, Nano Lett. **18**, 434–441 (2018). Copyright 2018, ACS]. (d) TEM cross-section with damage to the MoS₂ active layer caused by the deposition of the top electrode highlighted [Reproduced with permission from Huang *et al.*, Adv. Funct. Mater. **34**, 2214250 (2024). Copyright 2024, Wiley-VCH] and schematic illustrating the use of defect engineering to improve the reliability of MoS₂ [Reproduced with permission from Wu *et al.*, npj 2D Mater Appl **6**, 31 (2022). Copyright 2022, Springer⁶⁴]. FE memories: (e) HfO₂ ferroelectricity origin; yellow and red represent Hf and O atoms, respectively. The transition of the m- to t- and polar o-phases due to dopant size, capping, and strain [Reproduced with permission from Banerjee, Kashir, and Kamba, Small **18**, 2107575 (2022). Copyright 2022 Wiley]. P-V hysteresis loop for FE HZO capacitors (Hf:Zr 1:1) at varying thicknesses [Reproduced with permission from Zhou *et al.*, Nat. Commun. **15**, 2893 (2024). Copyright 2024, Springer⁶⁵]. (f) Sr-doped HfO₂ FE capacitors' P_r evolution with bipolar cycling. Difference between +P_r and -P_r determines memory window [Reproduced with permission from Wu *et al.*, npj 2D Mater. Appl. **6**, 31 (2022). Copyright 2022, Springer⁶⁴].

sufficient understanding of the material structure to identify failure modes. The conductive mechanism in oxide-materials-based resistive switching devices is complex. The switching behavior often has been reported to involve interactions among defects,⁹⁴

electrons,⁹⁵ metal ions,⁹⁶ interfaces,⁹⁷ and even grain boundaries.⁹⁸ In the published literature, it is challenging to examine an atomic device and fully elucidate the physical mechanisms of resistance switching. The unregulated movement of ions inside the memristor

TABLE I. Performance parameters from prominent oxide-based memristors.

Material	TE/BE	V _{SET} /V _{RESET} (V)	R _{OFF} /R _{ON}	Retention	Endurance	References
ZnO	Ag/Cu	1.2/−1.25	1000	...	>500 cycles	66
ZnO	Pt/Pt	1/−0.5	100	...	10 ⁶ cycles	67
TiO ₂	Pt/Pt	...	100	...	...	68
TiO ₂	TiN/Pt	+1/−1.5	10	10 ⁴ s	10 ⁴ cycles	69
NiO	Pt/Pt	>10/<−10	...	> 10 ⁴ s	...	70
NiO	Au/Au	+5.2/−6	...	...	...	71
HfO ₂	TiN/TiN	1.5/−1.4	100	>500 min at 200 °C	> 10 ⁶ cycles	72
HfO _x	TiN/TiN	...	>50	10 ⁵ s at 200 °C	5 × 10 ⁷ cycles	73
ZrO ₂	ITO/Ag	1/−1	>10	10 ⁶ s at 27 °C	>50 cycles	74
ZrO ₂	TiN/Pt	0.8/−0.5	...	10 ⁴ s at 27 °C	10 ³ cycles	75
CeO ₂	Au/Au	2.4/−3	104	...	...	8
AlO _x	Cu/W	1.3/−0.05	500	10 ³ s	...	76
Al ₂ O ₃	Ti/Pt	1.4/−1.7	<1000	10 ⁴ s	...	77
Gd ₂ O ₃	ITO/ITO	+2/−2	...	...	10 ³ cycles	78
GdO _x	Cr/TiN	<+4/−4	>70	3 × 10 ⁴ s	10 ⁵ cycles	79

is a critical component contributing to its performance deficiencies [Fig. 2(b)]. In contrast to conventional MOS transistors, which depend on the movement of a large number of electrons, memristors depend on the migration of a restricted number of ions via a designated layer or interface. The restricted number of ions, along with the stochastic characteristics of their movement and the morphology of the resultant ion migration region, complicates the precise regulation and quantification of the memristor’s performance, resulting in randomness and inconsistency.⁹² Sophisticated characterization techniques capable of elucidating switching pathways are needed to understand the device kinetics at the atomic scale. This is imperative toward the development of ultrathin switching memristor devices.

B. Two-dimensional NVMeS

The discovery of graphene in 2004 paved the way for explosive growth in the field of 2D materials.³³ Advantageous characteristics of 2D materials include strong in-plane covalent bonding, which provides mechanical strength while maintaining flexibility, fast switching speeds, low power consumption, and excellent device scalability.^{99–107} Using monolayers enables the formation of devices that are at the physical limit of minimum thickness, maximizing the potential for high-density integration. For example, where conductive filaments one atom wide are formed in a monolayer-based memristor [Fig. 2(c)], the term “atomristor” is sometimes used. A significant benefit is the potential for an atomically flat electrode contact, enabled by the absence of dangling bonds on the surface of a van der Waals material. The use of 2D materials facilitates the circumvention of scalability challenges associated with reducing bulk oxides to similar thicknesses. Nonetheless, achieving satisfactory levels of memory and endurance now constitutes an obstacle to scaling up.

Among various 2D materials proposed over the years, graphene is a mechanically strong yet flexible VdW electrical conductor

with the highest electrical conductivity of any known material at room temperature. It has an electron mobility of up to $2 \times 10^5 \text{ cm}^2/\text{Vs}$,¹⁰⁸ but this is strongly affected by the fabrication process.¹⁰⁹ The monolayer shows optical transparency to white light of 97.7%.¹¹⁰ These properties make graphene highly appropriate for transparent conductive electrodes, and competitive with the standard oxide electrode indium tin oxide (ITO). The ideal graphene lattice is a zero-band gap semimetal,¹¹¹ with modification of the electronic band structure achieved by the introduction of point defects, wrinkles and crumples, dislocations, or grain boundaries.¹¹² These many degrees of freedom for controlling the electronic structure of graphene offer flexibility to optimize the material properties according to the device application.^{113,114} Graphene is therefore commonly used as an electrode material in 2D materials-based memristive devices and is also combined with a secondary material as a composite to form the active switching layer.^{115–123} In the years immediately following the discovery of graphene, graphene oxide was explored as a 2D insulating dielectric for use in NVMe devices.^{34–37} Subsequently, hBN has emerged as the preeminent VdW material used as an insulator and is used to perform a range of functions in 2D electronics,^{124,125} including as a switching layer for ReRAM with ultralow energy consumption.^{118–120,126–139} However, the dielectric constant of hBN is relatively low, and fabrication challenges lead to the presence of defect states in the bandgap, which lead to undesirable tunnel currents in few-layer hBN,^{140,141} so other 2D insulators are sought after. Candidate materials include bismuth oxyhalides BiOX (X = Br, I),^{142–144} Bi₂SeO₅,¹⁴⁵ and quasi-2D ultrathin crystalline CaF₂.¹⁴⁶

Transition metal dichalcogenides (TMDs) are a class of semiconducting VdW materials with a trilayer structure utilized as transistor channels or switching layers in 2D NVMe devices.¹⁴⁷ Various semiconductive VdW materials, including Group IV Monochalcogenides (SnS¹⁴⁸ and GeSe¹⁴⁹), 2D III-V semiconductors (GaSe and InSe^{150–152}), ultrathin metal oxides (MoO₃ and TiO_x,^{153–155}), black phosphorus,^{156–158} and organic–inorganic quasi-2D halide

TABLE II. Examples of 2D materials used for NVM application.

Materials	TE/BE	$V_{\text{SET}}/V_{\text{RESET}}$	$R_{\text{OFF}}/R_{\text{ON}}$	Retention	Endurance	References
Graphene oxide	Al/Al	$-2.5/+2.5$ V	100	10^5 s	100 cycles	37
hBN	Au/Au	$+3/-1$	10^5-10^7	6×10^5 s	50 cycles	139
BiOBr	Pt/Pt	$+3/-2$	10^5	10^4 s	280 cycles	142
Black phosphorus	Mg/W	...	10^3	...	250	158
Black phosphorus/Al ₂ O ₃ /black phosphorus	Ni/Au	...	10^6	...	5×10^3	197
GaSe	Ag/Ag	$-1/+1.5$	10^3	10^4	5000	198
MoS ₂	Au/Au	1.8/1.1	5	...	...	177
MoS _{2-x} O _x s	Graphene/graphene	3.5/-4.8	100	10^5 s	2×10^7	199
MoTe ₂ , Al ₂ O ₃	TE (Ti/Ni) BE (Ti/Au)	2.9/-1.75 V	10^5-10^6	...	...	171
ReS ₂	Ti/Ti	0/5	50	700 s	>100	165
CuInP ₂ S ₆	Cr/graphene	$-5.5/+4.5$	10^7	10^4 s	5000 cycles	200
SnS ₂ , hBN	Cr/Au	40/-40	$\sim 2.58 \times 10^8$	2.4×10^5 s	15 000 cycles	167
In ₂ Se ₃	Cr/graphene	$+0.7/+3$	50	1200	100 cycles	201

perovskites.¹⁵⁹⁻¹⁶¹ 2D TMDs (NbTe₄^{162,163} ReS₂,¹⁶⁴⁻¹⁶⁶ SnS₂,¹⁶⁷ MoTe₂,¹⁶⁸⁻¹⁷¹ and MoS₂^{116,121,135,172-183}) currently dominate discussion of industry-relevant 2D semiconductors.^{184,185} MoS₂ is highly suitable for use as the active layer in memristors, due to high carrier mobility, well-established fabrication methods, robust Young's modulus, direct bandgap, and chemical stability under ambient conditions.¹⁸⁶⁻¹⁸⁸ The construction of NVMe based on ferroelectricity in 2D materials has been an object of recent interest.¹⁸⁹⁻¹⁹¹ Ferroelectric polarization does not normally persist for ultrathin oxides, although some successful preparations of ultrathin films of bulk oxides with preservation of ferroelectric behavior have been reported.¹⁹²⁻¹⁹⁴ Ferroelectric behavior in VdW materials has been reported for In₂Se₃,^{53-55,195} CuInP₂S₆,^{52,58,59} CuCrSe₂,⁵⁷ and γ -GeSe.⁵⁶ In-plane ferroelectric polarization has also been achieved in a single layer of SnTe⁵¹ and in few layers of SnS at room temperature.¹⁹⁶ Table II summarizes the performance metrics for some exemplar 2D memristive devices.

Low-power memristors are highly sought after for their potential in fast-speed, energy-efficient artificial neuromorphic networks. These devices are expected to consume less power than conventional CMOS devices due to their non-volatile nature and the absence of standby power dissipation.^{107,202} Unlike CMOS devices, which continuously dissipate power due to leakage currents, memristors only consume power during the switching process. Accurate power and energy calculations for memristors, however, require detailed knowledge of their I-V characteristics during switching, including parameters like threshold voltage, on-state current, and switching speed. Unfortunately, such data are not consistently available across all the memristors reported in the literature, thus not included in Table II.

Heterostructures in 2D memristor research improve switching behaviors, power consumption, and on/off ratios by customized interfaces and bandgap engineering.^{47,164,199,203} For example, it has been demonstrated that the addition of a hBN monolayer as a tunneling barrier reduces the electrode contact resistance for MoS₂-based devices.^{204,205} The design of these heterostructures is highly

flexible since the construction is based on VdW bonding, which eliminates the requirement for lattice matching.

There are a range of roadblocks to the widespread implementation of 2D materials in nano-electronic devices. A group of significant challenges is encountered in device fabrication. The growth and transfer of large-area, high-quality films are indeed challenges for all 2D material platforms, where fabrication compatible with current silicon chip technologies is particularly desirable.^{105,206} Damage to the 2D material can occur at various stages of device manufacture, including common crystallographic damage to the active layer during the deposition of the top electrode, as illustrated in Fig. 2(d). Since any damage to a 2D material is likely to impact a large fraction of the material volume, issues such as damage upon the deposition of the top electrode²⁰⁷ must be carefully accounted for. Stable and consistent doping of 2D materials for controlling device polarity is another prominent challenge.^{100,208,209} Another issue is incompatibility of 2D materials with currently widely available high-k dielectrics such as Al₂O₃ and HfO₂.^{100,210,211} The properties of 2D materials can be significantly impacted by the substrate, roughness, charged impurities, or the presence of any contaminants. This frequently requires dielectric encapsulation of 2D materials for stability, adding a further layer of complexity to device construction. The VdW insulator hBN may be employed for this purpose, but its suitability is limited by relatively low dielectric constant and by the difficulty of scaling up fabrication methods.²¹²

Given that 2D materials-based devices have thicknesses on the atomic scale, device performances are highly sensitive (evident from performance metrics listed in Table II) to sources of intrinsic and extrinsic disorder, such as crystallographic defects, interfacial Coulomb impurities, or fluctuations of the dielectric environment.²¹³⁻²¹⁵ Temporal and spatial variability is generally high in 2D materials-based NVMe devices, particularly in vertical devices based on filamentary switching. The probabilistic nature of the formation and fracture of the filament means that the threshold voltages follow a stochastic distribution, which can have a particularly large width for 2D materials. Where the variability is very

high, this property has been harnessed to mimic spiking action in neurons and for encryption purposes.^{116,158} Figure 2(d) illustrates one such approach used in a study that employed electron irradiation to increase the number of sulfur vacancies in a monolayer of MoS₂, thereby improving both device yield and endurance.⁶⁴ While non-filamentary mechanisms have also been demonstrated, such as voltage-bias-induced motion of sulfur vacancies in ReS₂-based lateral memristor,¹⁶⁵ this issue of variability must be addressed for proper utilization of vertical filamentary 2D materials based memristive technologies. This can only be achieved with complete characterization of local structure, dynamics, and sources of disorder during operation, to identify device mechanisms and design for optimized performance accordingly.

C. Ferroelectric NVMeS

Ferroelectric NVMeS operate on the principle of the use of electric polarization orientation for encoding information. The thermodynamic properties of ferroelectrics can be explained by Landau theory,^{216,217} which assumes that the ferroelectric phase (*lower symmetric phase*) transition occurs through a small distortion of the structure of the paraelectric phase (*higher symmetric phase*). Figure 2(e) depicts the influence of strain via the substrate and top electrode (capping effect) on the development of ferroelectricity in HfO₂. As the thickness of the deposited film diminishes, smaller grains develop post-annealing. This influences the energy landscape of the film, perhaps stabilizing the ferroelectric orthorhombic phase. Ultrathin films may benefit from elevated surface energy, which promotes higher-symmetry phases. The application of stress during manufacture may result in the development of ferroelectric phases. In contrast, thicker films (about 20 nm) are inclined to develop the non-polar monoclinic phase. The best thickness for achieving superior ferroelectric characteristics in HfO₂-based thin films is around 5–10 nm, striking a balance between surface energy advantages and the reduction of detrimental interfacial layer effects.²⁵ Ferroelectric memories offer a promising future for in-memory computing and energy-efficient miniature devices to meet computation-intensive processes. These devices have especially shown big promise in low power consumption and fast read/write speed.²¹⁸ Several fluorite- and perovskite-based oxide materials have been investigated, showing variability in device performance and reliability in terms of endurance (Table III).

There are, however, several challenges such as high coercive field, wake-up, and fatigue that are blocking the large-scale commercialization of these devices; some of those are discussed below.

Thin-film ferroelectricity observed in HfO₂ and ZrO₂ led to a paradigm shift from conventional perovskites (PZT, BTO) in realizing extremely scaled down NVMeS.²⁸ A major impediment in realizing this has, however, been high coercive fields observed in the former, which are essential to achieve low voltage/low power consumption. The higher E_c value indicates the higher kinetic energy barrier between the two polarization states,²²³ which is reported to be the kinetic energy that can switch *o*-HfO₂ by displacing oxygen ions along the polar axis.²⁵ The study of the possible intermediate phases was also conducted by Maeda *et al.*,²²⁴ who established that there are two intermediate states in the spontaneous switching pathway, *o*-phase (*Pbcm*) and *t*-phase (*P42/nmc*), with the lowest energy pathways seeming the most plausible. The energy barrier for the switching pathway involving the *o*-phase (*Pbcm*) at zero polarization was about twice as high (200 meV/f.u.) compared to that of the *t*-phase (*P42/nmc*) at zero polarization (75 meV/f.u.). Consequently, the pathway involving the *t*-phase (*P42/nmc*) is considered the most viable route for polarization switching in polar *o*-HfO₂.^{225–227} A direct experimental in-operando verification of these events is limited by phenomenological and instrumental limitations. A verification of phase transformation at several points in the polarization switching pathway will go a long way in having a deeper understanding of the coercive field problems.

While the reduction in the coercive field remains a challenge, an increase or decrease in polarization charge over several thousand read/write cycles introduces unreliability. The gradual increase in P_r upon the application of a cycling electric field is referred to as the wake-up effect.²²⁸ This is a primary difficulty hindering the large-scale commercializing of HfO₂-based ferroelectric devices. In published literature, two processes are identified as probable reasons behind the wake-up effect; the electric field-induced non-polar to polar phase transition and domain wall pinning and depinning influenced by point defects, particularly oxygen vacancies.²²⁹ The STEM analysis [Fig. 2(f)] of the HfO₂ film indicated a phase transition from a non-polar *t*- or *m*-phase to a polar *o*-phase with the application of an electric field.²³⁰ The transition, corroborated by high-angle annular dark-field (HAADF)-STEM analysis in Fig. 2(f), accounts for the observed rise in P_r . This phase change may be a direct consequence of applied bias or may result from the redistribution of oxygen vacancies under the influence of a strong electric field.²³⁰ The second prominent reason is anticipated to be point defects, particularly oxygen vacancies.^{30,231,232} A direct investigation of these phenomena requires direct mapping of atomistic processes inside the ultrathin functional layer under applied bias and essentially tracking under ambient conditions to isolate stray effects.

TABLE III. Common classes of FE materials used for NVM applications.

Material	P_r ($\mu\text{C cm}^{-2}$)	E_c (kV cm^{-1})	Thickness (nm)	ϵ_{FE}	E_c/E_{bd} (%)	References
Pb (Zr,Ti)O ₃	10–40	50–70	≈25	≈200	≈3.5–5.0	219
SrBi ₂ Ta ₂ O ₉	5–10	30–50	≈35	≈150	≈2.5–4.1	220
(Hf,Zr)O ₂	10–40	800–2000	<5	≈30	≈16–40	221
(Al,Sc)N	80–110	2000–5000	5	≈9	≈33–83	222

Like wake-up, another significant issue limiting the commercialization of fluorite-based oxide NVMe is the fatigue phenomenon. The remnant polarization in ferroelectric thin films diminishes after $\sim 10^5$ cycles [Fig. 2(f)], hence progressively narrowing the memory window and leading to device failure. Various mechanisms are hypothesized to produce fatigue in different types of ferroelectrics during electric field cycling.^{233,234} The growth of a non-polar and low permittivity (ϵ_r) layer between the functional electrode and the FE thin film has been reported in several studies.²³⁴ A different process proposed relies on the buildup of oxygen vacancies.^{235,236} Oxygen vacancies were reported to move and accumulate at the film/electrode interfaces and film grain boundaries during field cycling, resulting in domain-wall pinning. The buildup of oxygen vacancies at the grain boundaries, which restrict domain walls and reduce polarization switching in the presence of an applied electric field, can lead to a gradual decline in the P_r value.^{235,236} Consequently, the development of large grains throughout the fabrication process could potentially elevate the number of cycles until fatigue occurs.^{237,238}

III. STATE OF THE ART: CURRENT CHARACTERIZATION TECHNIQUES

We have seen how challenges, including high variability, damage during fabrication, and sensitivity to disorder in 2D materials; high coercive field, wake-up, and fatigue in ferroelectrics; the challenge of balancing retention and power requirements in filamentary devices; and poor spatial and temporal uniformity, as well as low endurance, are all the result of a lack of understanding of materials and atomic dynamics. Therefore, characterization methods become essential tools for the development of new NVM technologies as they unravel intrinsic material characteristics, device functionalities, and failure causes. Advanced microscopy, spectroscopy, and *in situ* electrical measurements may offer comprehensive insights into atomic-scale phenomena, with the aim to optimize material interfaces, develop dependable switching mechanisms, and enhance device reliability, thus expediting the incorporation of novel memory solutions into practical applications.

Electron microscopy (EM) is often utilized to analyze the surface and internal structure of devices at the level of atomic structure. EM techniques can reach sub-nm resolution and yield multidimensional datasets, including tomography, diffraction patterns, and elemental analysis, but also come with a set of stringent limitations.²³⁹ Scanning Electron Microscopy (SEM) collects information about the topography and composition, with a spatial resolution in the range of 0.5–5 nm.²⁴⁰ SEM on Fe-doped STO shows structural changes in an Au/Fe:STO/Nb:STO memristive device²⁴¹ revealing the appearance of a feature marked as the “forming crater,” formed following a current spike during [Fig. 3(a)]. Higher magnification of the contrast spot and subsequent XPS points to Sr enrichment and Ti^{3+} and Fe^{2+} ions in the developing crater; crucial to generating conductive filaments. In a similar study, filament formation dynamics have been observed for electrochemical metallization (ECM) memory Pt/H₂O/Ag, with the cell in a lateral geometry to facilitate the use of SEM [Fig. 3(a)].²⁴² The conclusions drawn about the interaction of the morphology of the Ag dendrites with the switching characteristics of the system are limited by the discrepancy between the vacuum environment of the SEM and the ambient conditions under

which the devices are generally operated, as well as the requirement for lateral geometry rather than the usual vertical three-dimensional structure of memristors.

Preparing samples for transmission electron microscopy (TEM) requires resource-intensive methods to achieve the required sample thickness and cleanliness, meaning TEM has high costs in terms of both time and money (Fig. 3). In an *ex situ* study on FE HfO₂,²³⁴ TEM was utilized to analyze Gd:HfO₂ film structural changes at electrode interfaces during pristine, wake-up, and fatigue cycling conditions. The bulk monoclinic and tetragonal (both paraelectric) symmetry relaxed at the electrode interfaces toward the orthorhombic (ferroelectric) phase as per TEM results [Fig. 3(b)]. This relaxation trend was most pronounced in the pristine state and least pronounced in the fatigued state. This shows that the wake-up effect is caused by interface phase transformations from non-ferroelectric to ferroelectric, increasing switchable material volume. With sustained cycling, TEM observations of interface relaxation decreasing coincide with fatigue, suggesting a relationship between switchable areas at interfaces and leftover polarization. In memristive devices, TEM revealed filament dynamics in ZnO-based resistive memory. Researchers employed *in situ* TEM to study conducting filament development, confirming their conical shape and elemental Zn composition.²⁴³ As the applied voltage increases, the CF, seen by TEM, extends across the ZnO layer, finally establishing a connection between the top and bottom electrodes [Fig. 3(c)]. The study clearly demonstrated that the SET process initiates from a residual filament left behind from the previous reset process. TEM also showed filament breakdown at the inert electrode contact during erasing. The capacity to directly observe filaments and their behavior at various phases of operation revealed filament development dynamics. It has been highlighted that observations made under the vacuum conditions of EM may differ from the characteristics under the devices' normal operating conditions, with the presence of atmosphere having strong influence on the electrochemical behavior of ferroelectric and other devices.^{244,245} For example, RESET failure in HfO₂-based CBRAM was found to occur under vacuum and was attributed to a dual CF mechanism associated with moisture desorption, an effect not present in atmosphere.²⁴⁶

Scanning Tunneling Microscopy (STM) provides exceptional spatial resolution and is utilized for both visualization and modification of surfaces at the atomic level. It may be considered a less complex method than EM, with less time required per device measurement but still significant financial investment required (Fig. 3). STM characterization of MoS₂ nanoclusters [Fig. 3(d)] helps comprehend their atomic-scale electrical characteristics.²⁴⁷ Specifically, STM imaging and spectroscopy show one-dimensional metallic edge states in MoS₂ nanoclusters. The conductive states at the margins of MoS₂ nanoclusters are caused by electronic structural changes, not dangling bonds. Localized edge states provide bright features at the edges of MoS₂ clusters in STM pictures. The Fermi level is pinned 1.2 eV above the valence band maximum in these states. These metallic edge states may increase material conductivity, particularly in nanoscale devices with a high edge-to-bulk ratio. When a 2D material approaches the nanoribbon limit, edge states as observed via STM are projected to dominate electron conductance. However, STM is limited to the study of the top surface of electrically conducting materials and is generally performed under high-vacuum and

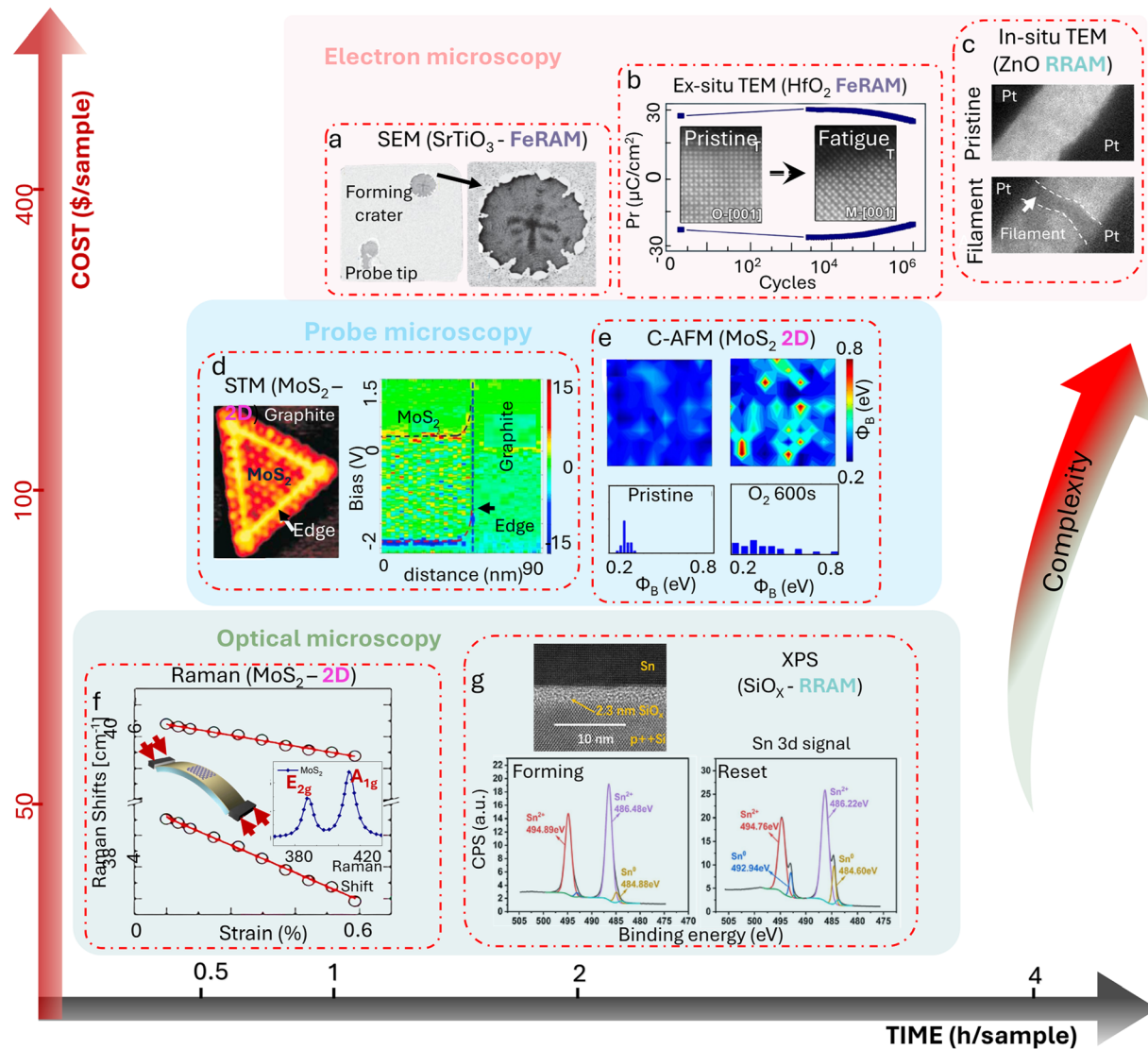


FIG. 3. Electron microscopy: (a) SEM image of CF dendrite in a lateral $\text{Ag}/\text{H}_2\text{O}/\text{Pt}$ resistive switch. [Reproduced with permission from Guo *et al.*, Appl. Phys. Lett. **91**, 133513 (2007). Copyright 2007 Wiley-VCH]. (b) HAADF-STEM images of doped HfO_2 in pristine and fatigued conditions reveal bulk m- (PE) and t- (PE) symmetry relaxing toward o- (FE) phase at electrode interfaces, most evident in pristine and least pronounced in fatigued [Reproduced with permission from Pešić *et al.*, Adv. Funct. Mater. **26**, 4601–4612 (2016). Copyright 2016 Wiley-VCH]. (c) TEM images of a ZnO-based planar device in pristine state (top) and post-forming (bottom). Arrows highlight typical filaments [Reproduced with permission from Chen *et al.*, Nano Lett. **13**, 3671–3677 (2013). Copyright 2013 ACS]. Probe microscopy: (d) An atomic-resolution STM image ($41 \times 42 \text{ \AA}^2$) of a MoS_2 nanocluster (left), depicting (right) the band profile of the monolayer MoS_2 and graphite interface, acquired in terms of $d[\log(dI/dV)]/dV$. The derivative of the logarithmic conductance with respect to V gives a measure of how quickly the logarithmic conductance changes. The spatial delineations of the valence band maximum and conduction band maximum are shown by the red and black dashed lines, respectively [Reproduced with permission from S. M. Hus and A.-P. Li, Prog. Surf. Sci. **92**, 176–201 (2017). Copyright 2017 Elsevier]. (e) Schottky barrier height maps and histograms for pristine MoS_2 (top left and bottom left) and after O_2 plasma treatment for 600 s (top right and bottom right) [Reproduced with permission from Giannazzo *et al.*, Nanomaterials **10**, 803 (2020). Copyright 2020 ACS]. Optical microscopy: (f) Measurement of MoS_2 bending and its effect on Raman shift [Reproduced with permission from Lee *et al.*, Nanoscale **15**, 7227–7248 (2023). Copyright 2023 RSC]. (g) Sn 3d signals collected from the active layer of a p-Si/ SiO_x /Sn device after cycling and etching of TE. Changes to XPS spectra show an increase in Sn^0 peaks after reset as compared to after forming, indicating a reduction of Sn ions. [Reproduced and adapted with permission from Chen *et al.*, Appl. Surf. Sci. **625**, 157191 (2023). Copyright 2023 Elsevier].

low-temperature conditions, limiting its applicability to the study of material dynamics within NVMe devices.

Conductive Atomic Force Microscopy (C-AFM) blends high-resolution topographical imaging with local electrical characterization, with intermediate cost and time requirements. It detects current flow through a material under an applied voltage using a conductive AFM tip to explore nanoscale electrical characteristics.²⁴⁸ This method helps study local conductivity, resistance, and charge transport processes in semiconductor devices, thin films, and organic electronics. C-AFM on MoS₂ measures the uniformity of the Schottky barrier height (SBH) at metal–MoS₂ device interfaces. SBH determines the sort of charge carriers (electrons for n-type, holes for p-type) that may enter TMD-based transistor channels. Figure 3(e) illustrates how oxygen plasma treatment adjusts the SBH of MoS₂, enabling low SBH areas for electrons and holes.²⁴⁹ This discovery enables the creation of ambipolar MoS₂ transistors and 2D memristors, a vital step in creating complementary MOS technology. While silicon-based CMOS technology requires both n- and p-type silicon, complementary MOS technology requires both types of MoS₂.

X-ray techniques are also employed to provide information about surface composition and chemical states. The application of bias during XPS experiments allows for *in situ* study of how electron binding energies vary during cycling.²⁵⁰ On the other hand, Fig. 3(g) illustrates the XPS spectra collected *ex situ* to investigate the resistive switching mechanism in a p-Si/SiO_x/Sn device.²⁵¹ Spectra were collected from the active layer after forming and after reset processes, by etching away the top electrode. A comparison of the Sn⁰ peaks shows a significant increase after reset as compared to after forming, indicating a reduction of Sn ions, which the authors note supports an oxygen vacancy-supported mechanism of resistive switching. This study was limited to post-mortem analysis due to the requirement to strip the top electrode for access to the active layer. X-ray absorption spectroscopy (XAS) is a resource-heavy technique that generally uses synchrotron radiation to study local atomic and electronic structure. *In situ* XAS allows for a detailed study of compositional changes over the lifetime of a memristive device and has been used to identify ring-like patterns of oxygen migration as a contributing factor to device failure in tantalum oxide memristors.²⁵²

While other methods may be intrusive or destructive, Raman spectroscopy shines for being generally a non-invasive analytical method. As a tabletop method performed under ambient conditions with no special sample preparation requirements, Raman is low-complexity and low-cost while still yielding rich real-time data. This unique position relative to other operando characterization methods is illustrated in Fig. 3. Measuring the inelastic scattering of monochromatic light, usually from a laser, yields comprehensive information on molecular structure, chemical composition, and interactions. Figure 3(f) tackles the problem of establishing Grüneisen characteristics by studying strain-phonon frequency relationships for 2D materials like MoS₂.²⁵³ MoS₂ bending data are shown in Fig. 3(f) employing a flexible cruciform substrate. Researchers can correctly assess strain-induced vibrational property changes by comparing Raman shifts with applied strain.

Although the techniques discussed above are all beneficial, none of them can be considered a universal solution. In fact, there

is a significant demand to be able to characterize and observe two-dimensional materials and thin films in their natural/ambient environment. In addition, there is a need to concurrently evaluate the topography, electrical characteristics, and spectroscopic behavior of these materials.

IV. NOVEL CHARACTERIZATION METHODS: FUTURE OF ENERGY MATERIALS INVESTIGATION

NVMe devices are generally based on interactions between defects,⁹⁴ electrons,⁹⁵ metal ions,⁹⁶ interfaces,⁹⁷ and grain boundaries.⁹⁸ Efficient characterization of NVMe devices should examine the correlation of these dynamic interactions with electrical characteristics in real time and under normal operating conditions. The rapid device development and iteration demanded by the current market pace means that characterization must also be fast, scalable, repeatable, and high-throughput. Measurement techniques based on electron microscopy or those requiring the use of a synchrotron x-ray source cannot be considered to meet these requirements. Scanning probe techniques face challenges including probe-sample interactions, tip reproducibility, and electrode durability. Electronic characterization approaches are often the principal research focus; however, although they provide extensive information on device performance characteristics, they often fail to provide crucial insights into the nano-kinetics inside the functional material, which are typically obscured by electrical measurements alone. Therefore, novel *operando* microscopy methods are emerging as highly effective approaches for rapidly developing an understanding of functional materials. Non-perturbative methods based on optical and plasmonic phenomena offer a unique route to understanding device dynamics.^{254,255}

Photoluminescent (PL) emissions are widely used to characterize semiconductors, yielding information about the electronic structure, including insight into any defect energy levels. Measurement of real-time variation in these optical signals offers a powerful *in situ* approach to real-time tracking of material modifications. Operando Raman spectroscopy can be used to detect real-time symmetry changes in the active medium of a device during electrical switching. For example, an operando Raman study of a thin-film niobium oxide has demonstrated that sodium (de) intercalation drives interfacial resistive switching, a factor relevant to device lifetime.³ Operando Raman and PL have been used to evidence activation of sodium cations in Na-doped WO₃ resistive switching devices.²⁵⁶

Plasmon (the resonant oscillation of electrons within a material when illuminated by light at the resonance frequency)²⁵⁷ phenomena can boost the capability to read out optical signals and have been used to enhance signal detection due to high field confinement within active devices, as well as to gather information about the materials where they are confined, as the plasmonic resonant frequency strongly depends on the materials composition and device morphology.^{254,255,258} A key advantage of the plasmonic enhancement of optical emissions is the ability to collect information from a nano-volume of material. The construction of resistive switches in a “nanoparticle-on-mirror” (NPoM) geometry has been shown to generate a tightly confined plasmonic hotspot whose volume overlaps with the switching material. This generates an optical system highly sensitive to local changes within the switching material,

resulting in a non-destructive tool for rapid device characterization under ambient conditions. Morphological changes due to electrical cycling of a nanoscale electrochemical metallization (ECM)-based switch in NPoM geometry have been detected via the modulation of light scattered by the nano junction. The formation of CFs caused a change in the overall geometry of the optical nano cavity (coinciding with the switching gap), hence modifying the plasmonic

resonance condition.²⁵⁹ The same approach has also been used in valence change memories (VCM), where the change of refractive index due to accumulation of oxygen vacancies in strontium titanate film has also been detected via frequency shift of plasmonic resonances.²⁶⁰

Plasmonic junctions have also been leveraged to combine plasmon-enhanced dark field (DF) nano-spectroscopy, nano-

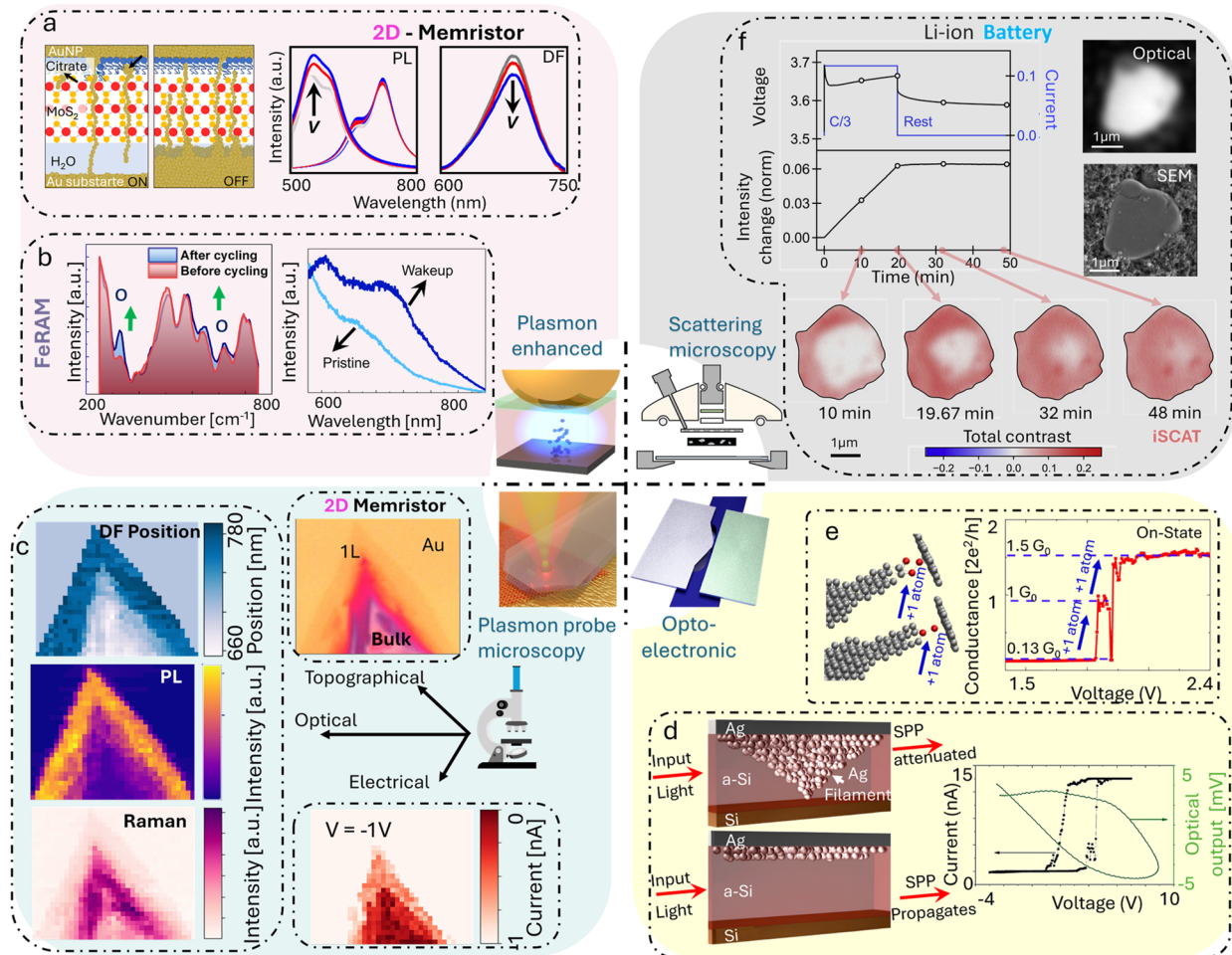


FIG. 4. Plasmon-enhanced: (a) Atomic-size Au filaments penetrate MoS₂ modifying resistivity, with citrate ligands hindering Au migration and creating stochastic "semi-nonvolatility" (black arrows). Gap resonance (DF) diminishes with increasing voltage. Photoluminescence spectra at 5, 0, and -5 V are averaged over several switching cycles [Reproduced with permission from Symonowicz *et al.*, *Adv. Mater.* **35**, 2209968 (2023). Copyright 2023 Wiley-VCH¹⁷⁴]. (b) SERS spectrum (left) of AuNP/HZO/TiN device before (blue) and after (red) cycling, showing ΔQ progression across 10^3 cycles. PL (right) spectra of the device before wake-up (blue) and pristine (cyan), with relative polarization charge (ΔQ) progression after 10^3 cycles [Reproduced with permission from Jan *et al.*, *Adv. Funct. Mater.* **33**, 2214970 (2023). Copyright 2023 Wiley-VCH]. Plasmon probe microscopy: (c) Combined optoelectrical characterization of MoS₂ using SPM displays a microscope image of the mapped flake and current measurement, a blue shift of the dark-field resonant gap mode peak from 1L to bulk MoS₂, and Raman and PL spectra [Reproduced with permission from Symonowicz *et al.*, *ACS Nano* **18**, 20412–20421 (2024). Copyright 2024 ACS]. Opto-electronic: (d) Formation and destruction of the nanoscale filament in the on-state and off-state (Left). The differential signal as a function of voltage illustrates the optical switching capabilities of the optical RRAM (right) [Reproduced with permission from Emboras *et al.*, *Nano Lett.* **13**, 6151–6155 (2013). Copyright 2013 ACS]. (e) Indication of switching using atomic-scale displacements. A conductance plot (red curve) illustrates quantized values resulting from a current-voltage sweep [Reproduced with permission from Emboras *et al.*, *Nano Lett.* **16**, 709–714 (2016). Copyright 2016 ACS]. Scattering microscopy: (f) The top subplot shows voltage and current profiles (black and blue) during 20 min of charging followed by a rest period. The bottom subplot shows iSCAT intensity change averaged across the active particle. At the bottom, normalized differential images of the active particle are shown for the times indicated [Reproduced with permission from Xu *et al.*, *Joule* **6**, 2535–2546 (2022). Copyright 2022 Elsevier].

Raman, and nano-PL to examine the volatile switching of a device based on MoS₂.¹⁷⁴ With applied voltage, the Au PL increase [Fig. 4(a)] suggests metal intercalation into MoS₂, while the sulfur vacancy-associated PL peak at 740 nm shows no change. Contrary to earlier theories, this clearly shows that sulfur vacancies do not participate in switching and that Au atoms are intercalating between Mo and S atoms, without modifying the MoS₂ structure. The change in plasmonic gap mode intensity [Fig. 4(a)] has been able to prove [supported by finite-difference time-domain (FDTD) simulations] the creation of many Au nanofilaments rather than a single one. The possibility to optically “count” the number of single Au nanofilaments per cycle event unveiled the key role of citrate ligands present at the interface of the top electrode of the MoS₂ device switch. In fact, ligands had a “semi-memristive” action that slows Au atom movement and prevents Au from retracting back to the Au electrode, this way contributing to device nonvolatility.²⁶¹ This is a clear example on how operando optical tracking of device switching contributed to give pivotal insights on the switching mechanism, opening surface engineering as a new route toward MoS₂-based switches.

The use of a combination of operando DF, operando PL, and operando Raman has been illustrated by a range of studies.³¹ In fact, excitation of PL-enhanced emissions allows interrogation of the defect states in the active material within the hotspot volume. As a result, the local defect dynamics during electrical switching can be examined. For example, a study of AuNP/HZO/TiN devices connected “wake-up” and “fatigue” effects to oxygen vacancy (Vo) migration.³⁰ Highly localized crystallographic data collected by surface-enhanced Raman spectroscopy (SERS) indicated a reduction in the monoclinic phase and a rise in the orthorhombic/tetragonal phases [Fig. 4(b)]. Interrogation of defect states by photoluminescence (PL) data showed a rise in oxygen vacancies in the film during a “pre-wake-up” stage, but a return to a defect-free film level with continued cycling, implying that the Vo buildup became more uniformly diffused. Hence, the operando optical data indicated that a crystallographic phase transition caused by migration of Vo was the source of the “wake-up” in these devices.

Clearly, plasmonic nanocavities have been evidenced as a powerful investigative technique for the *in situ* study of device dynamics. To make this approach technologically relevant, statistics should be gathered from multiple such devices for thorough characterization of the material. This can be time consuming. This bottleneck has been circumvented by attaching a plasmonically active top electrode to a portable conductive cantilever, free to be moved across the material of interest to apply bias at any desired location. This opto-electrical technique is termed Scanning Plasmon-Enhanced Microscopy (SPEM) and has been demonstrated on 2D-TMDs MoS₂ and WSe₂.²⁶² SPEM analyzes nanomaterials simultaneously using topographical, electrical, and spectroscopic methods, unlike previous scanning probe methods. Gold nanoparticles’ van der Waals interaction permits precise electrical characterization without harming the tested materials. Electrical and optical nanoscale mapping of MoS₂ (conductivity, dark-field, Raman, and photoluminescence) had been performed simultaneously in a study where MoS₂’s optoelectronic capabilities were observed to change with thickness²⁶² [Fig. 4(c)]. SPEM’s plasmonic hotspot boosted Raman and PL signals, enabling higher-resolution mapping than laser mapping. SPEM enhanced spectral resolution from 2.2 μm (without plasmonic

enhancement) to 600 nm. This improved resolution reveals small structural differences that affect MoS₂’s optical and electrical characteristics. Dark-field measurements at several points on the flakes were utilized to analyze MoS₂’s local optical characteristics using SPEM. In the field of NVMe, the aim is to have a facile method to understand how topographical or crystallographic features such as defects, domains, or sample edges may influence the local switching characteristics of the material. This is facilitated by SPEM’s ability to simultaneously collect electrical and optical information that is both spatially and temporally resolved.

Modulation of plasmonic resonances can be used as a mechanism for optical readout of memristor switching. For example, state switching of an RRAM integrated with a plasmonic waveguide has been detected via the disruption of SPP propagation by the formation of the metallic conductive filament [Fig. 4(d)].²⁶³ In another interesting work,²⁶⁴ an electrically controlled plasmonic switch functioning at the atomic scale was developed. The switch enables rapid and repeatable transitions by the repositioning of a single atom or, at most, a few atoms inside a plasmonic cavity. This study discusses the development of an atomic-scale plasmonic switch exhibiting step-like switching capability, as seen in Fig. 4(e). A fundamental issue in comprehending memristors resides in monitoring conductance quantization and conductance fingerprints, which reveal functioning at the atomic level. This work employs a method that fabricates an integrated atomic-scale optical switch on a silicon-on-insulator wafer. The researchers developed almost planarized waveguides using a modified local oxidation of silicon (LOCOS) technique. They then constructed metallic pads of silver and platinum, filling the interstice between the two metals with amorphous silicon (a-Si). An atomic-scale filament structure was ultimately created by applying a voltage, which facilitated the movement of silver ions to form a filamentary channel. This filament, serving as a plasmonic cavity, is modulated by moving a single or a few atoms, resulting in either a conductive or non-conductive condition that varies the plasmonic cavity characteristics, leading to unique plasmonic resonances. This procedure allows the observation of conductance quantization, signifying the effective functioning of a memristor at the atomic scale. Furthermore, where the atomic filament is formed due to light-induced thermal changes, the memristive switching can be used as a mechanism of photodetection.²⁶⁵

Novel techniques that have been employed for the *in situ* study of other devices such as batteries may also have potential applications in the study of NVMe. To tackle the difficulty of comprehending dynamic lithium-ion intercalation processes in battery materials, which constrains advancements in battery technology, particularly rapid charging, researchers used interferometric scattering microscopy (iSCAT). Figure 4(f) illustrates the application of this technique for tracing lithium-ion heterogeneity within a nickel-rich manganese cobalt oxide (NMC) cathode. This laboratory-based, cost-effective method has been used to trace lithium-ion heterogeneity within a nickel-rich manganese cobalt oxide (NMC) cathode,²⁶⁶ with iSCAT intensity increasing (or decreasing) during de-lithiation or (lithiation). Lithium distribution during charging was closely examined [Fig. 3(f)] and found to follow a “core-shell” pattern, with de-lithiation occurring at the particle periphery first. When the particle was allowed to rest after charging, a more uniform distribution was obtained, as shown in the intensity maps of the

NMC particle in Fig. 3(f). These observations were found to be consistent with kinetically-driven intra-particle heterogeneity during charging and allowed the authors to estimate the lithium-ion diffusion coefficient for the cathode material. iSCAT has also been utilized to track lithium-ion dynamics in Li_xCoO_2 , the prototypical cathode material.²⁶⁷ The same intensity change upon lithiation was used to map the presence of propagating phase boundaries during the biphasic and lithium ordering transitions in Li_xCoO_2 . These results underscore the capability of iSCAT microscopy to provide insights at the nm-length scale into lithium-ion intercalation processes in battery materials under ambient/natural settings, with potential for extension to the study of ionic motion in memristive systems.^{268,269} The real-time tracing of the dynamics of ions has also been achieved by detecting perturbations of surface plasmon waves hosted by an inert plasmon optical fiber probe.²⁷⁰

V. CONCLUSION

Variability, reliability, and scalability continue to pose substantial difficulties in the commercialization of non-volatile memories (NVMs) like RRAM and FeRAM, which are considered among the most promising next-generation memory technologies. Two-dimensional materials offer a flexible platform for designing high-density, low-power RRAM, and FeRAM devices, but challenges arising from the lack of understanding of switching mechanisms are currently hindering progress. This study emphasizes the transformative capacity of advanced characterization techniques, such as plasmonic spectroscopy, nanoparticle-on-mirror (NPoM) structured nanoelectrodes for plasmon-enhanced mapping, interferometric scattering detection for ion migration, and state-of-the-art atomic-scale optoelectronic switches in tackling these challenges. Through an exhaustive examination of how these techniques elucidate the nanoscale intricacies of NVMs, we provide an innovative viewpoint on expediting the advancement and use of these essential technologies. The integration of traditional and novel characterization techniques can provide practical insights into material behavior with competitive precision, illuminating pathways to device optimization. These contributions provide a basis for forthcoming advancements in NVM technology and their extensive commercialization.

ACKNOWLEDGMENTS

G.D.M. acknowledges support from the EPSRC (Grant Nos. EP/R511675/1 and EP/X034593/1; selected by the ERC, funded by UKRI) and the Royal Society (Grant No. RGS/R1/221262). D.M.K. acknowledges support from the EPSRC (Grant No. EP/S022953/1) and the Cambridge Trust. All authors acknowledge support from the Winton Program for the Physics of Sustainability.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

A.J. and D.M.K. contributed equally to this work.

Atif Jan: Investigation (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Dawn M. Kelly:** Investigation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Giuliana Di Martino:** Conceptualization (lead); Funding acquisition (lead); Project administration (lead); Resources (lead); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

REFERENCES

- J. Shalf, “The future of computing beyond Moore’s Law,” *Philos. Trans. R. Soc. London, Ser. A* **378**(2166), 20190061 (2020).
- P. Gargini, F. Balestra, and Y. Hayashi, “Roadmapping of nanoelectronics for the new electronics industry,” *Appl. Sci.* **12**(1), 308 (2021).
- M. Coll, J. Fontcuberta, M. Althammer, M. Bibes, H. Boschker, A. Calleja, G. Cheng, M. Cuoco, R. Dittmann, B. Dkhil, I. El Baggari, M. Fanciulli, I. Fina, E. Fortunato, C. Frontera, S. Fujita, V. Garcia, S. T. B. Goennenwein, C.-G. Granqvist, J. Grollier, R. Gross, A. Hagfeldt, G. Herranz, K. Hono, E. Houwman, M. Huijben, A. Kalaboukhov, D. J. Keeble, G. Koster, L. F. Kourkoutis, J. Levy, M. Lira-Cantu, J. L. MacManus-Driscoll, J. Mannhart, R. Martins, S. Menzel, T. Mikolajick, M. Napari, M. D. Nguyen, G. Niklasson, C. Paillard, S. Panigrahi, G. Rijnders, F. Sánchez, P. Sanchis, S. Sanna, D. G. Schlom, U. Schroeder, K. M. Shen, A. Siemon, M. Spreitzer, H. Sukegawa, R. Tamayo, J. van den Brink, N. Pryds, and F. M. Granozio, “Towards oxide electronics: A Roadmap,” *Appl. Surf. Sci.* **482**, 1–93 (2019).
- M. Acosta, N. Novak, V. Rojas, S. Patel, R. Vaish, J. Koruza, G. A. Rossetti, and J. Rödel, “BaTiO₃-based piezoelectrics: Fundamentals, current status, and perspectives,” *Appl. Phys. Rev.* **4**(4), 041305 (2017).
- N. Izyumskaya, Y.-I. Alivov, S.-J. Cho, H. Morkoç, H. Lee, and Y.-S. Kang, “Processing, structure, properties, and applications of PZT thin films,” *Crit. Rev. Solid State Mater. Sci.* **32**(3–4), 111–202 (2007).
- J. H. Choi, Y. Mao, and J. P. Chang, “Development of hafnium based high-k materials—A review,” *Mater. Sci. Eng.: R Rep.* **72**(6), 97–136 (2011).
- J. J. Yang, D. B. Strukov, and D. R. Stewart, “Memristive devices for computing,” *Nat. Nanotechnol.* **8**(1), 13–24 (2013).
- L. Chua, “Memristor—The missing circuit element,” *IEEE Trans. Circuit Theory* **18**(5), 507–519 (1971).
- L. O. Chua and S. M. Kang, “Memristive devices and systems,” *Proc. IEEE* **64**(2), 209–223 (1976).
- S. Vongehr and X. Meng, “The missing memristor has not been found,” *Sci. Rep.* **5**(1), 11657 (2015).
- L. Chua, “Resistance switching memories are memristors,” *Appl. Phys. A* **102**(4), 765–783 (2011).
- T. W. Hickmott, “Low-frequency negative resistance in thin anodic oxide films,” *J. Appl. Phys.* **33**(9), 2669–2682 (1962).
- D. B. Strukov, G. S. Snider, D. R. Stewart, and R. S. Williams, “The missing memristor found,” *Nature* **453**(7191), 80–83 (2008).
- J. Valasek, “Piezo-electric and allied phenomena in Rochelle salt,” *Phys. Rev.* **17**(4), 475–481 (1921).
- D. A. Buck, “Ferroelectrics for digital information storage and switching,” MIT Digital Computer Laboratory **R-212**, 8 (1952).
- J. F. Scott, “Ferroelectric memories,” *Phys. World* **8**(2), 46 (1995).
- J. L. Moll and Y. Tarui, “A new solid state memory resistor,” *IEEE Trans. Electron Devices* **10**(5), 338 (1963).

- ¹⁸K. Asadi, M. Li, P. W. M. Blom, M. Kemerink, and D. M. de Leeuw, "Organic ferroelectric opto-electronic memories," *Mater. Today* **14**(12), 592–599 (2011).
- ¹⁹K. Sugibuchi, Y. Kurogi, and N. Endo, "Ferroelectric field-effect memory device using $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ film," *J. Appl. Phys.* **46**(7), 2877–2881 (1975).
- ²⁰J. F. Scott and C. A. Paz De Araujo, "Ferroelectric memories," *Science* **246**(4936), 1400–1405 (1989).
- ²¹D. Bondurant, "Ferroelectric ram memory family for critical data storage," *Ferroelectrics* **112**(1), 273–282 (1990).
- ²²T. S. Böscke, J. Müller, D. Bräuhäus, U. Schröder, and U. Böttger, "Ferroelectricity in hafnium oxide thin films," *Appl. Phys. Lett.* **99**(10), 102903 (2011).
- ²³J. Junquera and P. Ghosez, "Critical thickness for ferroelectricity in perovskite ultrathin films," *Nature* **422**(6931), 506–509 (2003).
- ²⁴D. J. Kim, J. Y. Jo, Y. S. Kim, Y. J. Chang, J. S. Lee, J.-G. Yoon, T. K. Song, and T. W. Noh, "Polarization relaxation induced by a depolarization field in ultrathin ferroelectric BaTiO_3 capacitors," *Phys. Rev. Lett.* **95**(23), 237602 (2005).
- ²⁵W. Banerjee, A. Kashir, and S. Kamba, "Hafnium oxide (HfO_2)—A multifunctional oxide: A review on the prospect and challenges of hafnium oxide in resistive switching and ferroelectric memories," *Small* **18**(23), 2107575 (2022).
- ²⁶H. Chen, X. Zhou, L. Tang, Y. Chen, H. Luo, X. Yuan, C. R. Bowen, and D. Zhang, " HfO_2 -based ferroelectrics: From enhancing performance, material design, to applications," *Appl. Phys. Rev.* **9**(1), 011307 (2022).
- ²⁷S. S. Das, Z. Fox, M. D. Mia, B. C. Samuels, R. Saha, R. Droopad, S. S. Das, Z. Fox, M. D. Mia, B. C. Samuels, R. Saha, and R. Droopad, "Demonstration of ferroelectricity in PLD grown HfO_2 - ZrO_2 nanolaminates," *AIMS Mater. Sci.* **10**(2), 342–355 (2023).
- ²⁸Z. Fan, J. Chen, and J. Wang, "Ferroelectric HfO_2 -based materials for next-generation ferroelectric memories," *J. Adv. Dielectr.* **06**(02), 1630003 (2016).
- ²⁹K. D. Kim, M. H. Park, H. J. Kim, Y. J. Kim, T. Moon, Y. H. Lee, S. D. Hyun, T. Gwon, and C. S. Hwang, "Ferroelectricity in undoped- HfO_2 thin films induced by deposition temperature control during atomic layer deposition," *J. Mater. Chem. C* **4**(28), 6864–6872 (2016).
- ³⁰A. Jan, T. Rembert, S. Taper, J. Symonowicz, N. Strkalj, T. Moon, Y. S. Lee, H. Bae, H. J. Lee, D.-H. Choe, J. Heo, J. MacManus-Driscoll, B. Monserrat, and G. Di Martino, "In operando optical tracking of oxygen vacancy migration and phase change in few nanometers ferroelectric HZO memories," *Adv. Funct. Mater.* **33**(22), 2214970 (2023).
- ³¹A. Jan, S. A. Fraser, T. Moon, Y. S. Lee, H. Bae, H. J. Lee, D.-H. Choe, M. T. Becker, J. L. MacManus-Driscoll, J. Heo, and G. Di Martino, "Resetting the drift of oxygen vacancies in ultrathin HZO ferroelectric memories by electrical pulse engineering," *Small Sci.* **4**, 2400223 (2024).
- ³²P. R. Wallace, "The band theory of graphite," *Phys. Rev.* **71**(9), 622–634 (1947).
- ³³K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, Y. Zhang, S. V. Dubonos, I. V. Grigorieva, and A. A. Firsov, "Electric field effect in atomically thin carbon films," *Science* **306**(5696), 666–669 (2004).
- ³⁴C. L. He, F. Zhuge, X. F. Zhou, M. Li, G. C. Zhou, Y. W. Liu, J. Z. Wang, B. Chen, W. J. Su, Z. P. Liu, Y. H. Wu, P. Cui, and R.-W. Li, "Nonvolatile resistive switching in graphene oxide thin films," *Appl. Phys. Lett.* **95**(23), 232101 (2009).
- ³⁵S. K. Hong, J. E. Kim, S. O. Kim, S.-Y. Choi, and B. J. Cho, "Flexible resistive switching memory device based on graphene oxide," *IEEE Electron Device Lett.* **31**(9), 1005–1007 (2010).
- ³⁶X. Zhuang, Y. Chen, G. Liu, P. Li, C. Zhu, E. Kang, K. Noeh, B. Zhang, J. Zhu, and Y. Li, "Conjugated-polymer-functionalized graphene oxide: Synthesis and nonvolatile rewritable memory effect," *Adv. Mater.* **22**(15), 1731–1735 (2010).
- ³⁷H. Y. Jeong, J. Y. Kim, J. W. Kim, J. O. Hwang, J.-E. Kim, J. Y. Lee, T. H. Yoon, B. J. Cho, S. O. Kim, R. S. Ruoff, and S.-Y. Choi, "Graphene oxide thin films for flexible nonvolatile memory applications," *Nano Lett.* **10**(11), 4381–4386 (2010).
- ³⁸B. Radisavljevic, A. Radenovic, J. Brivio, V. Giacometti, and A. Kis, "Single-layer MoS_2 transistors," *Nat. Nanotechnol.* **6**(3), 147–150 (2011).
- ³⁹J. Liu, Z. Zeng, X. Cao, G. Lu, L. Wang, Q. Fan, W. Huang, and H. Zhang, "Preparation of MoS_2 -polyvinylpyrrolidone nanocomposites for flexible non-volatile rewritable memory devices with reduced graphene oxide electrodes," *Small* **8**(22), 3517–3522 (2012).
- ⁴⁰A. A. Bessonov, M. N. Kirikova, D. I. Petukhov, M. Allen, T. Ryhänen, and M. J. A. Bailey, "Layered memristive and memcapacitive switches for printable electronics," *Nat. Mater.* **14**(2), 199–204 (2015).
- ⁴¹G. He, H. Ramamoorthy, C.-P. Kwan, Y.-H. Lee, J. Nathawat, R. Somphonsane, M. Matsunaga, A. Higuchi, T. Yamanaka, N. Aoki, Y. Gong, X. Zhang, R. Vajtai, P. M. Ajayan, and J. P. Bird, "Thermally assisted nonvolatile memory in monolayer MoS_2 transistors," *Nano Lett.* **16**(10), 6445–6451 (2016).
- ⁴²V. K. Sangwan, H.-S. Lee, H. Bergeron, I. Balla, M. E. Beck, K.-S. Chen, and M. C. Hersam, "Multi-terminal memtransistors from polycrystalline monolayer molybdenum disulfide," *Nature* **554**(7693), 500–504 (2018).
- ⁴³A. K. Geim and I. V. Grigorieva, "Van der Waals heterostructures," *Nature* **499**(7459), 419–425 (2013).
- ⁴⁴M. Sup Choi, G.-H. Lee, Y.-J. Yu, D.-Y. Lee, S. Hwan Lee, P. Kim, J. Hone, and W. Jong Yoo, "Controlled charge trapping by molybdenum disulfide and graphene in ultrathin heterostructured memory devices," *Nat. Commun.* **4**(1), 1624 (2013).
- ⁴⁵Y. Wang, E. Liu, A. Gao, T. Cao, M. Long, C. Pan, L. Zhang, J. Zeng, C. Wang, W. Hu, S.-J. Liang, and F. Miao, "Negative photoconductance in van der Waals heterostructure-based floating gate phototransistor," *ACS Nano* **12**(9), 9513–9520 (2018).
- ⁴⁶Q. Wang, Y. Wen, K. Cai, R. Cheng, L. Yin, Y. Zhang, J. Li, Z. Wang, F. Wang, F. Wang, T. A. Shifa, C. Jiang, H. Yang, and J. He, "Nonvolatile infrared memory in MoS_2/PbS van der Waals heterostructures," *Sci. Adv.* **4**(4), eaap7916 (2018).
- ⁴⁷R. Cheng, F. Wang, L. Yin, Z. Wang, Y. Wen, T. A. Shifa, and J. He, "High-performance, multifunctional devices based on asymmetric van der Waals heterostructures," *Nat. Electron.* **1**(6), 356–361 (2018).
- ⁴⁸J. Panisilvam, H. Y. Lee, S. Byun, D. Fan, and S. Kim, "Two-dimensional material-based memristive devices for alternative computing," *Nano Convergence* **11**(1), 25 (2024).
- ⁴⁹S. Li, M. Pam, Y. Li, L. Chen, Y. Chien, X. Fong, D. Chi, and K. Ang, "Wafer-scale 2D hafnium diselenide based memristor crossbar array for energy-efficient neural network hardware," *Adv. Mater.* **34**(25), 2103376 (2022).
- ⁵⁰H. Lee, V. K. Sangwan, W. A. G. Rojas, H. Bergeron, H. Y. Jeong, J. Yuan, K. Su, and M. C. Hersam, "Dual-gated MoS_2 memtransistor crossbar array," *Adv. Funct. Mater.* **30**(45), 2003683 (2020).
- ⁵¹K. Chang, J. Liu, H. Lin, N. Wang, K. Zhao, A. Zhang, F. Jin, Y. Zhong, X. Hu, W. Duan, Q. Zhang, L. Fu, Q.-K. Xue, X. Chen, and S.-H. Ji, "Discovery of robust in-plane ferroelectricity in atomic-thick SnTe ," *Science* **353**(6296), 274–278 (2016).
- ⁵²F. Liu, L. You, K. L. Seyler, X. Li, P. Yu, J. Lin, X. Wang, J. Zhou, H. Wang, H. He, S. T. Pantelides, W. Zhou, P. Sharma, X. Xu, P. M. Ajayan, J. Wang, and Z. Liu, "Room-temperature ferroelectricity in CuInP_2S_6 ultrathin flakes," *Nat. Commun.* **7**(1), 12357 (2016).
- ⁵³Y. Huang, N. Chen, Z. Li, X. Wang, H. Sun, S. Zhang, and X. Li, "Two-dimensional In_2Se_3 : A rising advanced material for ferroelectric data storage," *InfoMat* **4**(8), e12341 (2022).
- ⁵⁴P. Hou, Y. Lv, Y. Chen, Y. Liu, C. Wang, P. Zhou, X. Zhong, J. Wang, and X. Ouyang, "In-plane strain-modulated photoresponsivity of the α - In_2Se_3 -based flexible transistor," *ACS Appl. Electron. Mater.* **2**(1), 140–146 (2020).
- ⁵⁵S. M. Poh, S. J. R. Tan, H. Wang, P. Song, I. H. Abidi, X. Zhao, J. Dan, J. Chen, Z. Luo, S. J. Pennycook, A. H. Castro Neto, and K. P. Loh, "Molecular-beam epitaxy of two-dimensional In_2Se_3 and its giant electroresistance switching in ferroresistive memory junction," *Nano Lett.* **18**(10), 6340–6346 (2018).
- ⁵⁶C. Liu, S. Guan, H. Yin, W. Wan, Y. Wang, and Y. Zhang, " γ - GeSe : A two-dimensional ferroelectric material with doping-induced ferromagnetism," *Appl. Phys. Lett.* **115**(25), 252904 (2019).
- ⁵⁷Z. Sun, Y. Su, A. Zhi, Z. Gao, X. Han, K. Wu, L. Bao, Y. Huang, Y. Shi, X. Bai, P. Cheng, L. Chen, K. Wu, X. Tian, C. Wu, and B. Feng, "Evidence for multiferroicity in single-layer CuCrSe_2 ," *Nat. Commun.* **15**(1), 4252 (2024).
- ⁵⁸B. Li, S. Li, H. Wang, L. Chen, L. Liu, X. Feng, Y. Li, J. Chen, X. Gong, and K. Ang, "An electronic synapse based on 2D ferroelectric CuInP_2S_6 ," *Adv. Electron. Mater.* **6**(12), 2000760 (2020).
- ⁵⁹W. Ming, B. Huang, S. Zheng, Y. Bai, J. Wang, J. Wang, and J. Li, "Flexoelectric engineering of van der Waals ferroelectric CuInP_2S_6 ," *Sci. Adv.* **8**(33), eabq1232 (2022).
- ⁶⁰F. Qin, Y. Zhang, H. W. Song, and S. Lee, "Enhancing memristor fundamentals through instrumental characterization and understanding reliability issues," *Mater. Adv.* **4**(8), 1850–1875 (2023).

- ⁶¹X.-D. Li, N.-K. Chen, B.-Q. Wang, M. Niu, M. Xu, X. Miao, and X.-B. Li, "Resistive memory devices at the thinnest limit: Progress and challenges," *Adv. Mater.* **36**(15), 2307951 (2024).
- ⁶²X. Wu, S. Mei, M. Bosman, N. Raghavan, X. Zhang, D. Cha, K. Li, and K. L. Pey, "Evolution of filament formation in Ni/HfO₂/SiO_x/Si-based RRAM devices," *Adv. Electron. Mater.* **1**(11), 1500130 (2015).
- ⁶³K. C. Kwon, J. H. Baek, K. Hong, S. Y. Kim, and H. W. Jang, "Memristive devices based on two-dimensional transition metal chalcogenides for neuromorphic computing," *Nano-Micro Lett.* **14**(1), 58 (2022).
- ⁶⁴X. Wu, Y. Gu, R. Ge, M. I. Serna, Y. Huang, J. C. Lee, and D. Akinwande, "Electron irradiation-induced defects for reliability improvement in monolayer MoS₂-based conductive-point memory devices," *npj 2D Mater. Appl.* **6**(1), 31 (2022).
- ⁶⁵C. Zhou, L. Ma, Y. Feng, C.-Y. Kuo, Y.-C. Ku, C.-E. Liu, X. Cheng, J. Li, Y. Si, H. Huang, Y. Huang, H. Zhao, C.-F. Chang, S. Das, S. Liu, and Z. Chen, "Enhanced polarization switching characteristics of HfO₂ ultrathin films via acceptor-donor co-doping," *Nat. Commun.* **15**(1), 2893 (2024).
- ⁶⁶N. M. Muhammad, N. Duraisamy, K. Rahman, H. W. Dang, J. Jo, and K. H. Choi, "Fabrication of printed memory device having zinc-oxide active nano-layer and investigation of resistive switching," *Curr. Appl. Phys.* **13**(1), 90–96 (2013).
- ⁶⁷F.-C. Chiu, "Resistance switching characteristics in ZnO-based nonvolatile memory devices," *Adv. Mater. Sci. Eng.* **2013**(1), 362053.
- ⁶⁸J. Joshua Yang *et al.*, "Dopant control by atomic layer deposition in oxide films for memristive switches," *Chem. Mater.* **23**, 123 (2011).
- ⁶⁹Y. H. Do, J. S. Kwak, Y. C. Bae, J. H. Lee, Y. Kim, H. Im, and J. P. Hong, "TiN electrode-induced bipolar resistive switching of TiO₂ thin films," *Curr. Appl. Phys.* **10**(1), e71–e74 (2010).
- ⁷⁰K. Oka, T. Yanagida, K. Nagashima, M. Kanai, T. Kawai, J.-S. Kim, and B. H. Park, "Spatial nonuniformity in resistive-switching memory effects of NiO," *J. Am. Chem. Soc.* **133**(32), 12482–12485 (2011).
- ⁷¹S. Brivio, G. Tallarida, D. Perego, S. Franz, D. Deleruyelle, C. Muller, and S. Spiga, "Low-power resistive switching in Au/NiO/Au nanowire arrays," *Appl. Phys. Lett.* **101**(22), 223510 (2012).
- ⁷²H. Kim *et al.*, "Engineering chemically abrupt high-k metal oxide/silicon interfaces using an oxygen-gettering metal overlayer," *J. Appl. Phys.* **96**, 3467 (2004).
- ⁷³B. Govoreanu, G. S. Kar, Y.-Y. Chen, V. Paraschiv, S. Kubicek, A. Fantini, I. P. Radu, L. Goux, S. Clima, R. Degraeve, N. Jossart, O. Richard, T. Vandeweyer, K. Seo, P. Hendrickx, G. Pourtois, H. Bender, L. Altimime, D. J. Wouters, J. A. Kittl, and M. Jurczak, "10×10nm² Hf/HfO_x crossbar resistive RAM with excellent performance, reliability and low-energy operation," in *2011 International Electron Devices Meeting* (IEEE, 2011), pp. 31.6.1–31.6.4.
- ⁷⁴M. N. Awais, N. M. Muhammad, D. Navaneethan, H. C. Kim, J. Jo, and K. H. Choi, "Fabrication of ZrO₂ layer through electrohydrodynamic atomization for the printed resistive switch (memristor)," *Microelectron. Eng.* **103**, 167–172 (2013).
- ⁷⁵B. Sun, Y. X. Liu, L. F. Liu, N. Xu, Y. Wang, X. Y. Liu, R. Q. Han, and J. F. Kang, "Highly uniform resistive switching characteristics of TiN/ZrO₂/Pt memory devices," *J. Appl. Phys.* **105**(6), 061630 (2009).
- ⁷⁶A. Sleiman, P. W. Sayers, and M. F. Mabrook, "Mechanism of resistive switching in Cu/AlO_x/W nonvolatile memory structures," *J. Appl. Phys.* **113**(16), 164506 (2013).
- ⁷⁷C.-Y. Lin, C.-Y. Wu, C.-Y. Wu, C. Hu, and T.-Y. Tseng, "Bistable resistive switching in Al₂O₃ memory thin films," *J. Electrochem. Soc.* **154**(9), G189 (2007).
- ⁷⁸K.-C. Liu, W.-H. Tzeng, K.-M. Chang, Y.-C. Chan, and C.-C. Kuo, "Bipolar resistive switching effect in Gd₂O₃ films for transparent memory application," *Microelectron. Eng.* **88**(7), 1586–1589 (2011).
- ⁷⁹D. Jana, M. Dutta, S. Samanta, and S. Maikap, "RRAM characteristics using a new Cr/GdO_x/TiN structure," *Nanoscale Res. Lett.* **9**(1), 680 (2014).
- ⁸⁰X. Shan, C. Zhao, X. Wang, Z. Wang, S. Fu, Y. Lin, T. Zeng, X. Zhao, H. Xu, X. Zhang, and Y. Liu, "Plasmonic optoelectronic memristor enabling fully light-modulated synaptic plasticity for neuromorphic vision," *Adv. Sci.* **9**(6), 2104632 (2022).
- ⁸¹T. Deng, C. Ye, J. Wu, P. He, and H. Wang, "Improved performance of ITO/TiO₂/HfO₂/Pt random resistive accessory memory by nitrogen annealing treatment," *Microelectron. Reliab.* **57**, 34–38 (2016).
- ⁸²Y. Li, J. Chu, W. Duan, G. Cai, X. Fan, X. Wang, G. Wang, and Y. Pei, "Analog and digital bipolar resistive switching in solution-combustion-processed NiO memristor," *ACS Appl. Mater. Interfaces* **10**(29), 24598–24606 (2018).
- ⁸³Y. Lin, J. Liu, J. Shi, T. Zeng, X. Shan, Z. Wang, X. Zhao, H. Xu, and Y. Liu, "Nitrogen-induced ultralow power switching in flexible ZnO-based memristor for artificial synaptic learning," *Appl. Phys. Lett.* **118**(10), 103502 (2021).
- ⁸⁴L. Zhang, H. Huang, C. Ye, K.-C. Chang, R. Zhang, Q. Xia, X. Wei, W. Wei, and W. Wang, "Exploration of highly enhanced performance and resistive switching mechanism in hafnium doping ZnO memristive device," *Semicond. Sci. Technol.* **33**(8), 085013 (2018).
- ⁸⁵K.-J. Zhou, M.-C. Chen, T.-C. Chang, S.-K. Lin, Y.-B. Wang, Y.-C. Zhang, P.-Y. Yen, K.-Y. Shao, H.-C. Huang, J.-W. Huang, and S. M. Sze, "Interfacial variation in HfO₂-based resistive switching devices with titanium electrodes under asymmetric bias operation," *J. Phys. D: Appl. Phys.* **56**(6), 065101 (2023).
- ⁸⁶J. Shin, M. Kang, and S. Kim, "Gradual conductance modulation of Ti/WO₃/Pt memristor with self-rectification for a neuromorphic system," *Appl. Phys. Lett.* **119**(1), 012102 (2021).
- ⁸⁷C. Du, F. Cai, M. A. Zidan, W. Ma, S. H. Lee, and W. D. Lu, "Reservoir computing using dynamic memristors for temporal information processing," *Nat. Commun.* **8**(1), 2204 (2017).
- ⁸⁸P. He, C. Ye, J. Wu, W. Wei, X. Wei, H. Wang, R. Zhang, L. Zhang, Q. Xia, and H. Wang, "Effect of sputtering atmosphere on the characteristics of ZrO_x resistive switching memory," *Semicond. Sci. Technol.* **32**(5), 055016 (2017).
- ⁸⁹X. Wei, H. Huang, C. Ye, W. Wei, H. Zhou, Y. Chen, R. Zhang, L. Zhang, and Q. Xia, "Exploring the role of nitrogen incorporation in ZrO₂ resistive switching film for enhancing the device performance," *J. Alloys Compd.* **775**, 1301–1306 (2019).
- ⁹⁰M.-J. Lee, C. B. Lee, D. Lee, S. R. Lee, M. Chang, J. H. Hur, Y.-B. Kim, C.-J. Kim, D. H. Seo, S. Seo, U.-I. Chung, I.-K. Yoo, and K. Kim, "A fast, high-endurance and scalable non-volatile memory device made from asymmetric Ta₂O_{5-x}/TaO_{2-x} bilayer structures," *Nat. Mater.* **10**(8), 625–630 (2011).
- ⁹¹F. Zahoor, T. Z. Azni Zulkifli, and F. A. Khanday, "Resistive random access memory (RRAM): An overview of materials, switching mechanism, performance, multilevel cell (mlc) storage, modeling, and applications," *Nanoscale Res. Lett.* **15**(1), 90 (2020).
- ⁹²W. Sun, B. Gao, M. Chi, Q. Xia, J. J. Yang, H. Qian, and H. Wu, "Understanding memristive switching via in situ characterization and device modeling," *Nat. Commun.* **10**(1), 3453 (2019).
- ⁹³X. Zhao, J. Ma, X. Xiao, Q. Liu, L. Shao, D. Chen, S. Liu, J. Niu, X. Zhang, Y. Wang, R. Cao, W. Wang, Z. Di, H. Lv, S. Long, and M. Liu, "Breaking the current-retention dilemma in cation-based resistive switching devices utilizing graphene with controlled defects," *Adv. Mater.* **30**(14), 1705193 (2018).
- ⁹⁴W. Banerjee, Q. Liu, and H. Hwang, "Engineering of defects in resistive random access memory devices," *J. Appl. Phys.* **127**, 051101 (2020).
- ⁹⁵J. C. Slonczewski, "Current-driven excitation of magnetic multilayers," *J. Magn. Magn. Mater.* **159**(1–2), L1–L7 (1996).
- ⁹⁶S. E. Kim, M.-H. Kim, J. Jang, H. Kim, S. Kim, J. Jang, J.-H. Bae, I. M. Kang, and S.-H. Lee, "Systematic engineering of metal ion injection in memristors for complex neuromorphic computing with high energy efficiency," *Adv. Intell. Syst.* **4**(9), 2200110 (2022).
- ⁹⁷D. Wang, C. Zhang, C. Han, L. Qian, and X. Huang, "Interface engineering for a VO_x based memristor with an ultralow switching voltage and logic functions," *J. Alloys Compd.* **921**, 166226 (2022).
- ⁹⁸H. Dou, M. Hellenbrand, M. Xiao, Z. Hu, S. Kunwar, A. Chen, J. L. MacManus-Driscoll, Q. Jia, and H. Wang, "Engineering of grain boundaries in CeO₂ enabling tailorable resistive switching properties," *Adv. Electron. Mater.* **9**(5), 2201186 (2023).
- ⁹⁹S. Batool, M. Idrees, S.-R. Zhang, S.-T. Han, and Y. Zhou, "Novel charm of 2D materials engineering in memristor: When electronics encounter layered morphology," *Nanoscale Horiz.* **7**(5), 480–507 (2022).
- ¹⁰⁰D. Joksas, A. AlMutairi, O. Lee, M. Cubukcu, A. Lombardo, H. Kurebayashi, A. J. Kenyon, and A. Mehonic, "Memristive, spintronic, and 2D-materials-based devices to improve and complement computing hardware," *Adv. Intell. Syst.* **4**(8), 2200068 (2022).
- ¹⁰¹C. Liu, H. Chen, S. Wang, Q. Liu, Y.-G. Jiang, D. W. Zhang, M. Liu, and P. Zhou, "Two-dimensional materials for next-generation computing technologies," *Nat. Nanotechnol.* **15**(7), 545–557 (2020).

- ¹⁰²L. Yin, R. Cheng, Y. Wen, C. Liu, and J. He, "Emerging 2D memory devices for in-memory computing," *Adv. Mater.* **33**(29), 2007081 (2021).
- ¹⁰³J. Pan, Z. Wang, B. Zhao, J. Yin, P. Guo, Y. Yang, and T.-L. Ren, "Recent progress of non-volatile memory devices based on two-dimensional materials," *Chips* **3**(4), 271–295 (2024).
- ¹⁰⁴S. S. Teja Nibhanupudi, A. Roy, D. Veksler, M. Coupin, K. C. Matthews, M. Disiena, Ansh, J. V. Singh, I. R. Gearba-Dolocan, J. Warner, J. P. Kulkarni, G. Bersuker, and S. K. Banerjee, "Ultra-fast switching memristors based on two-dimensional materials," *Nat. Commun.* **15**(1), 2334 (2024).
- ¹⁰⁵T.-J. Ko, H. Li, S. A. Mofid, C. Yoo, E. Okogbue, S. S. Han, M. S. Shawkat, A. Krishnaprasad, M. M. Islam, D. Dev, Y. Shin, K. H. Oh, G.-H. Lee, T. Roy, and Y. Jung, "Two-dimensional near-atom-thickness materials for emerging neuromorphic devices and applications," *iScience* **23**(11), 101676 (2020).
- ¹⁰⁶H. Zhou, S. Li, K.-W. Ang, and Y.-W. Zhang, "Recent advances in in-memory computing: Exploring memristor and memtransistor arrays with 2D materials," *Nano-Micro Lett.* **16**(1), 121 (2024).
- ¹⁰⁷H. Duan, S. Cheng, L. Qin, X. Zhang, B. Xie, Y. Zhang, and W. Jie, "Low-power memristor based on two-dimensional materials," *J. Phys. Chem. Lett.* **13**(31), 7130–7138 (2022).
- ¹⁰⁸J.-H. Chen, C. Jang, S. Xiao, M. Ishigami, and M. S. Fuhrer, "Intrinsic and extrinsic performance limits of graphene devices on SiO₂," *Nat. Nanotechnol.* **3**(4), 206–209 (2008).
- ¹⁰⁹Y. Sun, M. Sun, and D. Xie, "Graphene electronic devices," in *Graphene* (Elsevier, 2018), pp. 103–155.
- ¹¹⁰R. R. Nair, P. Blake, A. N. Grigorenko, K. S. Novoselov, T. J. Booth, T. Stauber, N. M. R. Peres, and A. K. Geim, "Fine structure constant defines visual transparency of graphene," *Science* **320**(5881), 1308 (2008).
- ¹¹¹N. Tombros, C. Jozsa, M. Popinciuc, H. T. Jonkman, and B. J. Van Wees, "Electronic spin transport and spin precession in single graphene layers at room temperature," *Nature* **448**(7153), 571–574 (2007).
- ¹¹²G. Yang, L. Li, W. B. Lee, and M. C. Ng, "Structure of graphene and its disorders: A review," *Sci. Technol. Adv. Mater.* **19**(1), 613–648 (2018).
- ¹¹³B. Walters, M. V. Jacob, A. Amirsoleimani, and M. Rahimi Azghadi, "A review of graphene-based memristive neuromorphic devices and circuits," *Adv. Intell. Syst.* **5**(10), 2300136 (2023).
- ¹¹⁴F. Yang, Z. Liu, X. Ding, Y. Li, C. Wang, and G. Shen, "Carbon-based memristors for resistive random access memory and neuromorphic applications," *Chip* **3**(2), 100086 (2024).
- ¹¹⁵T. N. T. A. Aziz, A. B. Rosli, M. M. Yusoff, S. H. Herman, and Z. Zulkifli, "Transparent hybrid ZnO-graphene film for high stability switching behavior of memristor device," *Mater. Sci. Semicond. Process.* **89**, 68–76 (2019).
- ¹¹⁶H. Kalita, A. Krishnaprasad, N. Choudhary, S. Das, D. Dev, Y. Ding, L. Tetard, H.-S. Chung, Y. Jung, and T. Roy, "Artificial neuron using vertical MoS₂/graphene threshold switching memristors," *Sci. Rep.* **9**(1), 53 (2019).
- ¹¹⁷H. Zhao, Z. Dong, H. Tian, D. DiMarzi, M. Han, L. Zhang, X. Yan, F. Liu, L. Shen, S. Han, S. Cronin, W. Wu, J. Tice, J. Guo, and H. Wang, "Atomically thin femtojoule memristive device," *Adv. Mater.* **29**(47), 1703232 (2017).
- ¹¹⁸F. M. Puglisi, L. Larcher, C. Pan, N. Xiao, Y. Shi, F. Hui, and M. Lanza, "2D h-BN based RRAM devices," in *2016 IEEE International Electron Devices Meeting (IEDM)* (IEEE, San Francisco, CA, 2016), pp. 34.8.1–34.8.4.
- ¹¹⁹C. Pan, Y. Ji, N. Xiao, F. Hui, K. Tang, Y. Guo, X. Xie, F. M. Puglisi, L. Larcher, E. Miranda, L. Jiang, Y. Shi, I. Valov, P. C. McIntyre, R. Waser, and M. Lanza, "Coexistence of grain-boundaries-assisted bipolar and threshold resistive switching in multilayer hexagonal boron nitride," *Adv. Funct. Mater.* **27**(10), 1604811 (2017).
- ¹²⁰K. Qian, R. Y. Tay, M.-F. Lin, J. Chen, H. Li, J. Lin, J. Wang, G. Cai, V. C. Nguyen, E. H. T. Teo, T. Chen, and P. S. Lee, "Direct observation of indium conductive filaments in transparent, flexible, and transferable resistive switching memory," *ACS Nano* **11**(2), 1712–1718 (2017).
- ¹²¹S. M. Shinde, G. Kalita, and M. Tanemura, "Fabrication of poly(methyl methacrylate)-MoS₂/graphene heterostructure for memory device application," *J. Appl. Phys.* **116**(21), 214306 (2014).
- ¹²²W. Yuhang, W. Lingyuan, L. Guodong, and L. Lixin, "Non-destructive photo-voltaic reading of interface type memristors using graphene as transparent electrode," *J. Alloys Compd.* **740**, 273–277 (2018).
- ¹²³B. Yalagala, S. Khandelwal, D. J. and S. Badhulika, "Wirelessly destructible MgO-PVP-graphene composite based flexible transient memristor for security applications," *Mater. Sci. Semicond. Process.* **104**, 104673 (2019).
- ¹²⁴K. K. Kim, H. S. Lee, and Y. H. Lee, "Synthesis of hexagonal boron nitride heterostructures for 2D van der Waals electronics," *Chem. Soc. Rev.* **47**(16), 6342–6369 (2018).
- ¹²⁵Y. Gong, Z.-Q. Xu, D. Li, J. Zhang, I. Aharonovich, and Y. Zhang, "Two-dimensional hexagonal boron nitride for building next-generation energy-efficient devices," *ACS Energy Lett.* **6**(3), 985–996 (2021).
- ¹²⁶G. U. Siddiqui, M. M. Rehman, Y.-J. Yang, and K. H. Choi, "A two-dimensional hexagonal boron nitride/polymer nanocomposite for flexible resistive switching devices," *J. Mater. Chem. C* **5**(4), 862–871 (2017).
- ¹²⁷S. Chen, M. R. Mahmoodi, Y. Shi, C. Mahata, B. Yuan, X. Liang, C. Wen, F. Hui, D. Akinwande, D. B. Strukov, and M. Lanza, "Wafer-scale integration of two-dimensional materials in high-density memristive crossbar arrays for artificial neural networks," *Nat. Electron.* **3**(10), 638–645 (2020).
- ¹²⁸L. Sun, Y. Zhang, G. Han, G. Hwang, J. Jiang, B. Joo, K. Watanabe, T. Taniguchi, Y.-M. Kim, W. J. Yu, B.-S. Kong, R. Zhao, and H. Yang, "Self-selective van der Waals heterostructures for large scale memory array," *Nat. Commun.* **10**(1), 3161 (2019).
- ¹²⁹A. Ranjan, N. Raghavan, S. J. O'Shea, S. Mei, M. Bosman, K. Shubhakar, and K. L. Pey, "Conductive atomic force microscope study of bipolar and threshold resistive switching in 2D hexagonal boron nitride films," *Sci. Rep.* **8**(1), 2854 (2018).
- ¹³⁰L. Völkel, D. Braun, M. Belete, S. Kataria, T. Wahlbrink, K. Ran, K. Kistermann, J. Mayer, S. Menzel, A. Daus, and M. C. Lemme, "Resistive switching and current conduction mechanisms in hexagonal boron nitride threshold memristors with nickel electrodes," *Adv. Funct. Mater.* **34**(15), 2300428 (2024).
- ¹³¹W. Lin, P. Zhuang, D. Akinwande, X.-A. Zhang, and W. Cai, "Oxygen-assisted synthesis of hBN films for resistive random access memories," *Appl. Phys. Lett.* **115**(7), 073101 (2019).
- ¹³²Y. Shi, X. Liang, B. Yuan, V. Chen, H. Li, F. Hui, Z. Yu, F. Yuan, E. Pop, H.-S. P. Wong, and M. Lanza, "Electronic synapses made of layered two-dimensional materials," *Nat. Electron.* **1**(8), 458–465 (2018).
- ¹³³Y. Shi, C. Pan, V. Chen, N. Raghavan, K. L. Pey, F. M. Puglisi, E. Pop, H.-S. P. Wong, and M. Lanza, "Coexistence of volatile and non-volatile resistive switching in 2D h-BN based electronic synapses," in *2017 IEEE International Electron Devices Meeting (IEDM)* (IEEE, San Francisco, CA, 2017), pp. 5.4.1–5.4.4.
- ¹³⁴K. Qian, R. Y. Tay, V. C. Nguyen, J. Wang, G. Cai, T. Chen, E. H. T. Teo, and P. S. Lee, "Hexagonal boron nitride thin film for flexible resistive memory applications," *Adv. Funct. Mater.* **26**(13), 2176–2184 (2016).
- ¹³⁵H. Park, M. A. Mastro, M. J. Tadjer, and J. Kim, "Programmable multilevel memtransistors based on van der Waals heterostructures," *Adv. Electron. Mater.* **5**(10), 1900333 (2019).
- ¹³⁶X. Wu, R. Ge, P. Chen, H. Chou, Z. Zhang, Y. Zhang, S. Banerjee, M. Chiang, J. C. Lee, and D. Akinwande, "Thinnest nonvolatile memory based on monolayer h-BN," *Adv. Mater.* **31**(15), 1806790 (2019).
- ¹³⁷M. Kim, E. Pallicchi, R. Ge, X. Wu, G. Ducournau, J. C. Lee, H. Happy, and D. Akinwande, "Analogue switches made from boron nitride monolayers for application in 5G and terahertz communication systems," *Nat. Electron.* **3**(8), 479–485 (2020).
- ¹³⁸J. Ge, H. Huang, Z. Ma, W. Chen, X. Cao, H. Fang, J. Yan, Z. Liu, W. Wang, and S. Pan, "A sub-500 mV monolayer hexagonal boron nitride based memory device," *Mater. Des.* **198**, 109366 (2021).
- ¹³⁹F. Hui, C. Pan, Y. Shi, Y. Ji, E. Grustan-Gutierrez, and M. Lanza, "On the use of two dimensional hexagonal boron nitride as dielectric," *Microelectron. Eng.* **163**, 119–133 (2016).
- ¹⁴⁰S. Yang, K. Liu, Y. Xu, L. Liu, H. Li, and T. Zhai, "Gate dielectrics integration for 2D electronics: Challenges, advances, and outlook," *Adv. Mater.* **35**(18), 2207901 (2023).
- ¹⁴¹T. Knobloch, Y. Y. Illarionov, F. Ducry, C. Schleich, S. Wachter, K. Watanabe, T. Taniguchi, T. Mueller, M. Waltl, M. Lanza, M. I. Vexler, M. Luisier, and T. Grasser, "The performance limits of hexagonal boron nitride as an insulator for scaled CMOS devices based on two-dimensional materials," *Nat. Electron.* **4**(2), 98–108 (2021).

- ¹⁴²B. Xie, X. Zhang, S. Cheng, and W. Jie, "Resistive switching in 2D bismuth oxyhalide nanosheets for nonvolatile memory and emulation of leaky integrate-and-fire functions," *Mater. Des.* **222**, 111090 (2022).
- ¹⁴³W. Xie, Y. Zhong, D. Wang, L. Zhong, L. Han, Q. Yang, and W. Jie, "Co-existence of digital and analog resistive switching in 2D layered BiOI nanosheets for synaptic applications," *Mater. Des.* **234**, 112367 (2023).
- ¹⁴⁴S. R. Evans, K. A. Oyekun, R. Nam, M. R. Osanloo, E. Deylgat, S. Mansoori, S. Tiwari, M. V. Fischetti, H. Edwards, and W. G. Vandenberghe, "Two-dimensional oxyhalides for power electronics," *IEEE Trans. Mater. Electron Devices* **1**, 151–159 (2024).
- ¹⁴⁵Y. Y. Illarionov, T. Knobloch, and T. Grasser, "Native high-*k* oxides for 2D transistors," *Nat. Electron.* **3**(8), 442–443 (2020).
- ¹⁴⁶Y. Y. Illarionov, A. G. Banskichikov, T. Knobloch, D. K. Polyushkin, S. Wachter, V. V. Fedorov, S. M. Suturin, M. Stoger-Pollach, T. Mueller, M. I. Vexler, N. S. Sokolov, and T. Grasser, "Crystalline calcium fluoride: A record-thin insulator for nanoscale 2D electronics," in *2020 Device Research Conference (DRC)* (IEEE, Columbus, OH, 2020), pp. 1–2.
- ¹⁴⁷R. Ge, X. Wu, M. Kim, J. Shi, S. Sonde, L. Tao, Y. Zhang, J. C. Lee, and D. Akinwande, "Atomristor: Nonvolatile resistance switching in atomic sheets of transition metal dichalcogenides," *Nano Lett.* **18**(1), 434–441 (2018).
- ¹⁴⁸S. Sucharitakul, U. Rajesh Kumar, R. Sankar, F.-C. Chou, Y.-T. Chen, C. Wang, C. He, R. He, and X. P. A. Gao, "Screening limited switching performance of multilayer 2D semiconductor FETs: The case for SnS," *Nanoscale* **8**(45), 19050–19057 (2016).
- ¹⁴⁹M. Bi, X. Li, Z. Wang, R. Chen, L. Zhu, Z. Du, C. Zhang, Y. Du, and F. Wu, "Modulation of semiconductor doping of 2D GeSe in h-BN/GeSe van der Waals heterostructure," *J. Phys. Chem. C* **128**(43), 18550–18558 (2024).
- ¹⁵⁰H. Arora and A. Erbe, "Recent progress in contact, mobility, and encapsulation engineering of InSe and GaSe," *InfoMat* **3**(6), 662–693 (2021).
- ¹⁵¹Z. Yang and J. Hao, "Recent progress in 2D layered III–VI semiconductors and their heterostructures for optoelectronic device applications," *Adv. Mater. Technol.* **4**(8), 1900108 (2019).
- ¹⁵²H. Du, M. Tu, S. Luo, Y. Liu, X. Qiu, H. Lu, S. Li, S. Yuan, W. Huang, W. Jie, and J. Hao, "Reversible transition between bipolar resistive switching and threshold switching in 2D layered III–VI semiconductor GaSe," *Appl. Phys. Lett.* **116**(25), 253102 (2020).
- ¹⁵³B. A. Holler, K. Crowley, M. Berger, and X. P. A. Gao, "2D semiconductor transistors with Van der Waals oxide MoO₃ as integrated high- κ gate dielectric," *Adv. Electron. Mater.* **6**(10), 2000635 (2020).
- ¹⁵⁴Y. Hong, S. Lan, B. Pan, Z. Zhang, B. Chen, L. Zhang, and P. Wang, "Thinner 2D α -MoO₃ makes setting up memristors easier," *J. Materiomics* **10**(6), 1279–1289 (2024).
- ¹⁵⁵G. Cao, C. Gao, J. Wang, J. Lan, and X. Yan, "Memristor based on two-dimensional titania nanosheets for multi-level storage and information processing," *Nano Res.* **15**(9), 8419–8427 (2022).
- ¹⁵⁶Y. Zhou, D. Liu, J. Wang, Z. Cheng, L. Liu, N. Yang, Y. Liu, T. Xia, X. Liu, X. Zhang, C. Ye, Z. Xu, W. Xiong, P. K. Chu, and X.-F. Yu, "Black phosphorus based multicolor light-modulated transparent memristor with enhanced resistive switching performance," *ACS Appl. Mater. Interfaces* **12**(22), 25108–25114 (2020).
- ¹⁵⁷Y. Wang, F. Wu, X. Liu, J. Lin, J.-Y. Chen, W.-W. Wu, J. Wei, Y. Liu, Q. Liu, and L. Liao, "High on/off ratio black phosphorus based memristor with ultra-thin phosphorus oxide layer," *Appl. Phys. Lett.* **115**(19), 193503 (2019).
- ¹⁵⁸Y. Peng, N. Wang, R. Wang, Y. Cao, Y. Xin, S. Wang, T. Zeng, W. Zhang, X. Ma, Q. Feng, H. Wang, and Y. Hao, "Solution-processed black phosphorus film-based volatile memristor for encryption applications," *IEEE Electron Device Lett.* **44**(12), 2051–2054 (2023).
- ¹⁵⁹S. J. Kim, I. H. Im, J. H. Baek, S. Choi, S. H. Park, D. E. Lee, J. Y. Kim, S. Y. Kim, N.-G. Park, D. Lee, J. J. Yang, and H. W. Jang, "Linearly programmable two-dimensional halide perovskite memristor arrays for neuromorphic computing," *Nat. Nanotechnol.* **20**, 83 (2024).
- ¹⁶⁰B. Li, F. Xia, B. Du, S. Zhang, L. Xu, Q. Su, D. Zhang, and J. Yang, "2D halide perovskites for high-performance resistive switching memory and artificial synapse applications," *Adv. Sci.* **11**(23), 2310263 (2024).
- ¹⁶¹S. J. Kim, I. H. Im, J. H. Baek, S. H. Park, J. Y. Kim, J. J. Yang, and H. W. Jang, "Reliable and robust two-dimensional perovskite memristors for flexible-resistive random-access memory array," *ACS Nano* **18**(41), 28131–28141 (2024).
- ¹⁶²Y. Shuang, Q. Chen, M. Kim, Y. Wang, Y. Saito, S. Hatayama, P. Fons, D. Ando, M. Kubo, and Y. Sutou, "NbTe₄ phase-change material: Breaking the phase-change temperature balance in 2D van der Waals transition-metal binary chalcogenide," *Adv. Mater.* **35**(39), 2303646 (2023).
- ¹⁶³Y. Shuang, D. Ando, and Y. Sutou, "Conduction mechanism in amorphous NbTe₄ thin film," *Mater. Trans.* **65**(9), 1061–1066 (2024).
- ¹⁶⁴X. Xiong, J. Kang, Q. Hu, C. Gu, T. Gao, X. Li, and Y. Wu, "Reconfigurable logic-in-memory and multilingual artificial synapses based on 2D heterostructures," *Adv. Funct. Mater.* **30**(11), 1909645 (2020).
- ¹⁶⁵S. Li, B. Li, X. Feng, L. Chen, Y. Li, L. Huang, X. Fong, and K.-W. Ang, "Electron-beam-irradiated rhenium disulfide memristors with low variability for neuromorphic computing," *npj 2D Mater. Appl.* **5**(1), 1 (2021).
- ¹⁶⁶M. E. Pam, S. Li, T. Su, Y. Chien, Y. Li, Y. S. Ang, and K. Ang, "Interface-modulated resistive switching in Mo-irradiated ReS₂ for neuromorphic computing," *Adv. Mater.* **34**(30), 2202722 (2022).
- ¹⁶⁷M. A. Khan, S. Yim, S. Rehman, F. Ghafoor, H. Kim, H. Patil, M. F. Khan, and J. Eom, "Two-dimensional materials memory devices with floating metal gate for neuromorphic applications," *Mater. Today Adv.* **20**, 100438 (2023).
- ¹⁶⁸S. Hatayama, Y. Saito, K. Makino, N. Uchida, Y. Shuang, S. Mori, Y. Sutou, M. Krbal, and P. Fons, "Phase control of sputter-grown large-area MoTe₂ films by preferential sublimation of Te: Amorphous, 1T' and 2H phases," *J. Mater. Chem. C* **10**(29), 10627–10635 (2022).
- ¹⁶⁹B. Zhao, L. Xu, R. Peng, Z. Xin, R. Shi, Y. Wu, B. Wang, J. Chen, T. Pan, and K. Liu, "High-performance 2D ambipolar MoTe₂ lateral memristors by mild oxidation," *Small* **20**(43), 2402727 (2024).
- ¹⁷⁰B. Chen, J. Yao, J. Xia, R. Yang, and X. Miao, "A strain-sensitive flexible MoTe₂-based memristor for gesture recognition," *IEEE Electron Device Lett.* **44**(4), 622–625 (2023).
- ¹⁷¹F. Zhang, H. Zhang, S. Krylyuk, C. A. Milligan, Y. Zhu, D. Y. Zemlyanov, L. A. Bendersky, B. P. Burton, A. V. Davydov, and J. Appenzeller, "Electric-field induced structural transition in vertical MoTe₂- and Mo_{1-x}W_xTe₂-based resistive memories," *Nat. Mater.* **18**(1), 55–61 (2019).
- ¹⁷²M. Krbal, V. Prokop, A. A. Kononov, J. R. Pereira, J. Mistrik, A. V. Kolobov, P. J. Fons, Y. Saito, S. Hatayama, Y. Shuang, Y. Sutou, S. A. Rozhkov, J. R. Stellhorn, S. Hayakawa, I. Pis, and F. Bondino, "Amorphous-to-crystal transition in quasi-two-dimensional MoS₂: Implications for 2D electronic devices," *ACS Appl. Nano Mater.* **4**(9), 8834–8844 (2021).
- ¹⁷³X. Zhu, D. Li, X. Liang, and W. D. Lu, "Ionic modulation and ionic coupling effects in MoS₂ devices for neuromorphic computing," *Nat. Mater.* **18**(2), 141–148 (2019).
- ¹⁷⁴J. Symonowicz, D. Polyushkin, T. Mueller, and G. Di Martino, "Fully optical in operando investigation of ambient condition electrical switching in MoS₂ nanodevices," *Adv. Mater.* **35**(10), 2209968 (2023).
- ¹⁷⁵X. Feng, Y. Li, L. Wang, S. Chen, Z. G. Yu, W. C. Tan, N. Macadam, G. Hu, L. Huang, L. Chen, X. Gong, D. Chi, T. Hasan, A. V. Thean, Y. Zhang, and K. Ang, "A fully printed flexible MoS₂ memristive artificial synapse with femtojoule switching energy," *Adv. Electron. Mater.* **5**(12), 1900740 (2019).
- ¹⁷⁶B. Tang, H. Veluri, Y. Li, Z. G. Yu, M. Waqar, J. F. Leong, M. Sivan, E. Zamburg, Y.-W. Zhang, J. Wang, and A. V.-Y. Thean, "Wafer-scale solution-processed 2D material analog resistive memory array for memory-based computing," *Nat. Commun.* **13**(1), 3037 (2022).
- ¹⁷⁷S. M. Hus, R. Ge, P.-A. Chen, L. Liang, G. E. Donnelly, W. Ko, F. Huang, M.-H. Chiang, A.-P. Li, and D. Akinwande, "Observation of single-defect memristor in an MoS₂ atomic sheet," *Nat. Nanotechnol.* **16**(1), 58–62 (2021).
- ¹⁷⁸S. Yin, C. Song, Y. Sun, L. Qiao, B. Wang, Y. Sun, K. Liu, F. Pan, and X. Zhang, "Electric and light dual-gate tunable MoS₂ memristor," *ACS Appl. Mater. Interfaces* **11**(46), 43344–43350 (2019).
- ¹⁷⁹V. K. Sangwan, D. Jariwala, I. S. Kim, K.-S. Chen, T. J. Marks, L. J. Lauhon, and M. C. Hersam, "Gate-tunable memristive phenomena mediated by grain boundaries in single-layer MoS₂," *Nat. Nanotechnol.* **10**(5), 403–406 (2015).
- ¹⁸⁰P. Cheng, K. Sun, and Y. H. Hu, "Memristive behavior and ideal memristor of 1T phase MoS₂ nanosheets," *Nano Lett.* **16**(1), 572–576 (2016).

- ¹⁸¹S. Hao, X. Ji, F. Liu, S. Zhong, K. Y. Pang, K. G. Lim, T. C. Chong, and R. Zhao, "Monolayer MoS₂/WO₃ heterostructures with sulfur anion reservoirs as electronic synapses for neuromorphic computing," *ACS Appl. Nano Mater.* **4**(2), 1766–1775 (2021).
- ¹⁸²D. Li, B. Wu, X. Zhu, J. Wang, B. Ryu, W. D. Lu, W. Lu, and X. Liang, "MoS₂ memristors exhibiting variable switching characteristics toward biorealistic synaptic emulation," *ACS Nano* **12**(9), 9240–9252 (2018).
- ¹⁸³M. Kim, R. Ge, X. Wu, X. Lan, J. Tice, J. C. Lee, and D. Akinwande, "Zero-static power radio-frequency switches based on MoS₂ atomistors," *Nat. Commun.* **9**(1), 2524 (2018).
- ¹⁸⁴K. P. O'Brien, C. H. Naylor, C. Dorow, K. Maxey, A. V. Penumatcha, A. Vyatsikh, T. Zhong, A. Kitamura, S. Lee, C. Rogan, W. Mortelmans, M. S. Kavrik, R. Steinhardt, P. Buragohain, S. Dutta, T. Tronic, S. Clendenning, P. Fischer, E. S. Putna, M. Radosavljevic, M. Metz, and U. Avci, "Process integration and future outlook of 2D transistors," *Nat. Commun.* **14**(1), 6400 (2023).
- ¹⁸⁵K. S. Kim, J. Kwon, H. Ryu, C. Kim, H. Kim, E.-K. Lee, D. Lee, S. Seo, N. M. Han, J. M. Suh, J. Kim, M.-K. Song, S. Lee, M. Seol, and J. Kim, "The future of two-dimensional semiconductors beyond Moore's law," *Nat. Nanotechnol.* **19**(7), 895–906 (2024).
- ¹⁸⁶R. A. Wells and A. W. Robertson, "Molybdenum disulfide memristors for next generation memory and neuromorphic computing: Progress and prospects," *Adv. Electron. Mater.* **10**(10), 2400121 (2024).
- ¹⁸⁷S. Bertolazzi, J. Brivio, and A. Kis, "Stretching and breaking of ultrathin MoS₂," *ACS Nano* **5**(12), 9703–9709 (2011).
- ¹⁸⁸E. S. Kadantsev and P. Hawrylak, "Electronic structure of a single MoS₂ monolayer," *Solid State Commun.* **152**(10), 909–913 (2012).
- ¹⁸⁹Y. Joo, E. Hwang, H. Hong, S. Cho, and H. Yang, "Memory and synaptic devices based on emerging 2D ferroelectricity," *Adv. Electron. Mater.* **9**(8), 2300211 (2023).
- ¹⁹⁰H. Wang, Y. Wen, H. Zeng, Z. Xiong, Y. Tu, H. Zhu, R. Cheng, L. Yin, J. Jiang, B. Zhai, C. Liu, C. Shan, and J. He, "2D ferroic materials for nonvolatile memory applications," *Adv. Mater.* **36**, 2305044 (2024).
- ¹⁹¹X. Wang, C. Zhu, Y. Deng, R. Duan, J. Chen, Q. Zeng, J. Zhou, Q. Fu, L. You, S. Liu, J. H. Edgar, P. Yu, and Z. Liu, "Van der Waals engineering of ferroelectric heterostructures for long-retention memory," *Nat. Commun.* **12**(1), 1109 (2021).
- ¹⁹²S. R. Lee, L. Baasandorj, J. W. Chang, I. W. Hwang, J. R. Kim, J.-G. Kim, K.-T. Ko, S. B. Shim, M. W. Choi, M. You, C.-H. Yang, J. Kim, and J. Song, "First observation of ferroelectricity in ~1 nm ultrathin semiconducting BaTiO₃ films," *Nano Lett.* **19**(4), 2243–2250 (2019).
- ¹⁹³D. Ji, S. Cai, T. R. Paudel, H. Sun, C. Zhang, L. Han, Y. Wei, Y. Zang, M. Gu, Y. Zhang, W. Gao, H. Huan, W. Guo, D. Wu, Z. Gu, E. Y. Tsymlal, P. Wang, Y. Nie, and X. Pan, "Freestanding crystalline oxide perovskites down to the monolayer limit," *Nature* **570**(7759), 87–90 (2019).
- ¹⁹⁴D. D. Fong, G. B. Stephenson, S. K. Streiffer, J. A. Eastman, O. Auciello, P. H. Fuoss, and C. Thompson, "Ferroelectricity in ultrathin perovskite films," *Science* **304**(5677), 1650–1653 (2004).
- ¹⁹⁵S. Wan, Y. Li, W. Li, X. Mao, C. Wang, C. Chen, J. Dong, A. Nie, J. Xiang, Z. Liu, W. Zhu, and H. Zeng, "Nonvolatile ferroelectric memory effect in ultrathin α -In₂Se₃," *Adv. Funct. Mater.* **29**(20), 1808606 (2019).
- ¹⁹⁶N. Higashitarumizu, M. Kawamoto, C.-J. Lee, B.-H. Lin, F.-H. Chu, I. Yonemori, T. Nishimura, K. Wakabayashi, W.-H. Chang, and K. Nagashio, "Purely in-plane ferroelectricity in monolayer SnS at room temperature," *Nat. Commun.* **11**(1), 2428 (2020).
- ¹⁹⁷X. Xiong, X. Wang, Q. Hu, X. Li, and Y. Wu, "Flexible synaptic floating gate devices with dual electrical modulation based on ambipolar black phosphorus," *iScience* **25**(3), 103947 (2022).
- ¹⁹⁸Y. Yang, H. Du, Q. Xue, X. Wei, Z. Yang, C. Xu, D. Lin, W. Jie, and J. Hao, "Three-terminal memtransistors based on two-dimensional layered gallium selenide nanosheets for potential low-power electronics applications," *Nano Energy* **57**, 566–573 (2019).
- ¹⁹⁹M. Wang, S. Cai, C. Pan, C. Wang, X. Lian, Y. Zhuo, K. Xu, T. Cao, X. Pan, B. Wang, S.-J. Liang, J. J. Yang, P. Wang, and F. Miao, "Robust memristors based on layered two-dimensional materials," *Nat. Electron.* **1**(2), 130–136 (2018).
- ²⁰⁰J. Wu, H.-Y. Chen, N. Yang, J. Cao, X. Yan, F. Liu, Q. Sun, X. Ling, J. Guo, and H. Wang, "High tunnelling electroresistance in a ferroelectric van der Waals heterojunction via giant barrier height modulation," *Nat. Electron.* **3**(8), 466–472 (2020).
- ²⁰¹M. S. Choi, B. Cheong, C. H. Ra, S. Lee, J. Bae, S. Lee, G. Lee, C. Yang, J. Hone, and W. J. Yoo, "Electrically driven reversible phase changes in layered In₂Se₃ crystalline film," *Adv. Mater.* **29**(42), 1703568 (2017).
- ²⁰²D. S. Jeong, K. M. Kim, S. Kim, B. J. Choi, and C. S. Hwang, "Memristors for energy-efficient new computing paradigms," *Adv. Electron. Mater.* **2**(9), 1600090 (2016).
- ²⁰³S. H. Kim, M. U. Park, C. Lee, S.-G. Yi, M. Kim, Y. Choi, J. H. Cho, and K.-H. Yoo, "Rectifying optoelectronic memory based on WSe₂/graphene heterostructures," *Nanoscale Adv.* **3**(17), 4952–4960 (2021).
- ²⁰⁴J. Wang, Q. Yao, C. Huang, X. Zou, L. Liao, S. Chen, Z. Fan, K. Zhang, W. Wu, X. Xiao, C. Jiang, and W. Wu, "High mobility MoS₂ transistor with low Schottky barrier contact by using atomic thick h-BN as a tunneling layer," *Adv. Mater.* **28**(37), 8302–8308 (2016).
- ²⁰⁵X. Cui, E.-M. Shih, L. A. Jauregui, S. H. Chae, Y. D. Kim, B. Li, D. Seo, K. Pistunova, J. Yin, J.-H. Park, H.-J. Choi, Y. H. Lee, K. Watanabe, T. Taniguchi, P. Kim, C. R. Dean, and J. C. Hone, "Low-temperature ohmic contact to monolayer MoS₂ by van der Waals bonded Co/h-BN electrodes," *Nano Lett.* **17**(8), 4781–4786 (2017).
- ²⁰⁶L. Jia, J. Wu, Y. Zhang, Y. Qu, B. Jia, Z. Chen, and D. J. Moss, "Fabrication technologies for the on-chip integration of 2D materials," *Small Methods* **6**(3), 2101435 (2022).
- ²⁰⁷Y. Huang, Y. Gu, S. Mohan, A. Dolocan, N. D. Ignacio, S. Kutagulla, K. Matthews, A. Londoño-Calderon, Y. Chang, Y. Chen, J. H. Warner, M. T. Pettes, J. C. Lee, and D. Akinwande, "Reliability improvement and effective switching layer model of thin-film MoS₂ memristors," *Adv. Funct. Mater.* **34**(15), 2214250 (2024).
- ²⁰⁸Z. Wang, H. Xia, P. Wang, X. Zhou, C. Liu, Q. Zhang, F. Wang, M. Huang, S. Chen, P. Wu, Y. Chen, J. Ye, S. Huang, H. Yan, L. Gu, J. Miao, T. Li, X. Chen, W. Lu, P. Zhou, and W. Hu, "Controllable doping in 2D layered materials," *Adv. Mater.* **33**(48), 2104942 (2021).
- ²⁰⁹M. Chhowalla, D. Jena, and H. Zhang, "Two-dimensional semiconductors for transistors," *Nat. Rev. Mater.* **1**(11), 16052 (2016).
- ²¹⁰Y. Y. Illarionov, T. Knobloch, M. Jech, M. Lanza, D. Akinwande, M. I. Vexler, T. Mueller, M. C. Lemme, G. Fiori, F. Schwierz, and T. Grasser, "Insulators for 2D nanoelectronics: The gap to bridge," *Nat. Commun.* **11**(1), 3385 (2020).
- ²¹¹E. Fortunato, P. Barquinha, and R. Martins, "Oxide semiconductor thin-film transistors: A review of recent advances," *Adv. Mater.* **24**(22), 2945–2986 (2012).
- ²¹²C. R. Dean, A. F. Young, I. Meric, C. Lee, L. Wang, S. Sorgenfrei, K. Watanabe, T. Taniguchi, P. Kim, K. L. Shepard, and J. Hone, "Boron nitride substrates for high-quality graphene electronics," *Nat. Nanotechnol.* **5**(10), 722–726 (2010).
- ²¹³D. Rhodes, S. H. Chae, R. Ribeiro-Palau, and J. Hone, "Disorder in van der Waals heterostructures of 2D materials," *Nat. Mater.* **18**(6), 541–549 (2019).
- ²¹⁴A. Raja, L. Waldecker, J. Zipfel, Y. Cho, S. Brem, J. D. Ziegler, M. Kulig, T. Taniguchi, K. Watanabe, E. Malic, T. F. Heinz, T. C. Berkelbach, and A. Chernikov, "Dielectric disorder in two-dimensional materials," *Nat. Nanotechnol.* **14**(9), 832–837 (2019).
- ²¹⁵S.-L. Li, K. Tsukagoshi, E. Orgiu, and P. Samori, "Charge transport and mobility engineering in two-dimensional transition metal chalcogenide semiconductors," *Chem. Soc. Rev.* **45**(1), 118–151 (2016).
- ²¹⁶J. F. Webb, "On the physics of ferroelectrics," *Sci. Prog.* **86**(3), 203–234 (2003).
- ²¹⁷O. Auciello, J. F. Scott, and R. Ramesh, "The physics of ferroelectric memories," *Phys. Today* **51**(7), 22–27 (1998).
- ²¹⁸M. Hoffmann, A. J. Tan, N. Shanker, Y.-H. Liao, L.-C. Wang, J.-H. Bae, C. Hu, and S. Salahuddin, "Fast read-after-write and depolarization fields in high endurance n-type ferroelectric FETs," *IEEE Electron Device Lett.* **43**(5), 717–720 (2022).
- ²¹⁹D. H. Lee, Y. Lee, Y. H. Cho, H. Choi, S. H. Kim, and M. H. Park, "Unveiled ferroelectricity in well-known non-ferroelectric materials and their semiconductor applications," *Adv. Funct. Mater.* **33**(42), 2303956 (2023).
- ²²⁰C. A.-P. de Araujo, J. D. Cuchiaro, L. D. McMillan, M. C. Scott, and J. F. Scott, "Fatigue-free ferroelectric capacitors with platinum electrodes," *Nature* **374**(6523), 627–629 (1995).
- ²²¹U. Schroeder, C. S. Hwang, and H. Funakubo, *Ferroelectricity in Doped Hafnium Oxide: Materials, Properties and Devices* (Woodhead Publishing, 2019).

- ²²²D. Wang, J. Zheng, P. Musavigharavi, W. Zhu, A. C. Foucher, S. E. Trolier-McKinstry, E. A. Stach, and R. H. Olsson, "Ferroelectric switching in sub-20 nm aluminum scandium nitride thin films," *IEEE Electron Device Lett.* **41**(12), 1774–1777 (2020).
- ²²³M. H. Park, Y. H. Lee, T. Mikolajick, U. Schroeder, and C. S. Hwang, "Review and perspective on ferroelectric HfO₂-based thin films for memory applications," *MRS Commun.* **8**(3), 795–808 (2018).
- ²²⁴T. Maeda, B. Magyari-Kope, and Y. Nishi, "Identifying ferroelectric switching pathways in HfO₂: First principles calculations under electric fields," in *2017 IEEE International Memory Workshop (IMW)* (IEEE, 2017), pp. 1–4.
- ²²⁵D. H. Lee, Y. Lee, K. Yang, J. Y. Park, S. H. Kim, P. R. S. Reddy, M. Materano, H. Mulaosmanovic, T. Mikolajick, J. L. Jones, U. Schroeder, and M. H. Park, "Domains and domain dynamics in fluorite-structured ferroelectrics," *Appl. Phys. Rev.* **8**(2), 021312 (2021).
- ²²⁶S. V. Barabash, D. Pramanik, Y. Zhai, B. Magyari-Kope, and Y. Nishi, "Ferroelectric switching pathways and energetics in (Hf,Zr)O₂," *ECS Trans.* **75**(32), 107 (2017).
- ²²⁷M. H. Park, H. J. Kim, Y. H. Lee, Y. J. Kim, T. Moon, K. D. Kim, S. D. Hyun, and C. S. Hwang, "Two-step polarization switching mediated by a nonpolar intermediate phase in Hf_{0.4}Zr_{0.6}O₂ thin films," *Nanoscale* **8**(29), 13898–13907 (2016).
- ²²⁸P. Jiang, Q. Luo, X. Xu, T. Gong, P. Yuan, Y. Wang, Z. Gao, W. Wei, L. Tai, and H. Lv, "Wake-up effect in HfO₂-based ferroelectric films," *Adv. Electron. Mater.* **7**(1), 2000728 (2021).
- ²²⁹M. T. Becker, C. J. Burkhardt, R. Kleiner, and D. Koelle, "Impedance spectroscopy of ferroelectrics: The domain wall pinning element," *J. Appl. Phys.* **132**(4), 044104 (2022).
- ²³⁰M. Pešić, F. P. G. Fengler, L. Larcher, A. Padovani, T. Schenk, E. D. Grimley, X. Sang, J. M. LeBeau, S. Slesazeck, U. Schroeder, and T. Mikolajick, "Physical mechanisms behind the field-cycling behavior of HfO₂-based ferroelectric capacitors," *Adv. Funct. Mater.* **26**(25), 4601–4612 (2016).
- ²³¹Y.-J. Lin, C.-Y. Teng, S.-J. Chang, Y.-F. Liao, C. Hu, C.-J. Su, and Y.-C. Tseng, "Role of electrode-induced oxygen vacancies in regulating polarization wake-up in ferroelectric capacitors," *Appl. Surf. Sci.* **528**, 147014 (2020).
- ²³²S. Starschich, S. Menzel, and U. Böttger, "Evidence for oxygen vacancies movement during wake-up in ferroelectric hafnium oxide," *Appl. Phys. Lett.* **108**(3), 032903 (2016).
- ²³³A. Rakita, N. Nikolić, M. Mildner, J. Matiassek, and A. Elbe-Bürger, "Re-epithelialization and immune cell behaviour in an *ex vivo* human skin model," *Sci. Rep.* **10**(1), 1 (2020).
- ²³⁴X. J. Lou, M. Zhang, S. A. T. Redfern, and J. F. Scott, "Local phase decomposition as a cause of polarization fatigue in ferroelectric thin films," *Phys. Rev. Lett.* **97**(17), 177601 (2006).
- ²³⁵S. S. Fields, S. W. Smith, S. T. Jaszewski, T. Mimura, D. A. Dickie, G. Esteves, M. David Henry, S. L. Wolfley, P. S. Davids, and J. F. Ihlefeld, "Wake-up and fatigue mechanisms in ferroelectric Hf_{0.5}Zr_{0.5}O₂ films with symmetric RuO₂ electrodes," *J. Appl. Phys.* **130**(13), 134101 (2021).
- ²³⁶F. Huang, X. Chen, X. Liang, J. Qin, Y. Zhang, T. Huang, Z. Wang, B. Peng, P. Zhou, H. Lu, L. Zhang, L. Deng, M. Liu, Q. Liu, H. Tian, and L. Bi, "Fatigue mechanism of yttrium-doped hafnium oxide ferroelectric thin films fabricated by pulsed laser deposition," *Phys. Chem. Chem. Phys.* **19**(5), 3486–3497 (2017).
- ²³⁷X. Li, C. Li, Z. Xu, Y. Li, Y. Yang, H. Hu, Z. Jiang, J. Wang, J. Ren, C. Zheng, C. Lu, and Z. Wen, "Ferroelectric properties and polarization fatigue of La:HfO₂ thin-film capacitors," *Phys. Status Solidi RRL* **15**(4), 2000481 (2021).
- ²³⁸A. G. Chernikova, M. G. Kozodaev, D. V. Negrov, E. V. Korostylev, M. H. Park, U. Schroeder, C. S. Hwang, and A. M. Markeev, "Improved ferroelectric switching endurance of La-doped Hf_{0.5}Zr_{0.5}O₂ thin films," *ACS Appl. Mater. Interfaces* **10**(3), 2701–2708 (2018).
- ²³⁹S. Henning and R. Adhikari, "Scanning electron microscopy, ESEM, and X-ray microanalysis," in *Microscopy Methods in Nanomaterials Characterization* (Elsevier, 2017), pp. 1–30.
- ²⁴⁰A. E. Vladár and K. T. Arat, "Limits of resolutions in the scanning electron microscope," *Microsc. Microanal.* **29**(Supplement_1), 463–464 (2023).
- ²⁴¹C. Lenser, M. Patt, S. Menzel, A. Köhl, C. Wiemann, C. M. Schneider, R. Waser, and R. Dittmann, "Insights into nanoscale electrochemical reduction in a memristive oxide: The role of three-phase boundaries," *Adv. Funct. Mater.* **24**(28), 4466–4472 (2014).
- ²⁴²X. Guo, C. Schindler, S. Menzel, and R. Waser, "Understanding the switching-off mechanism in Ag⁺ migration based resistively switching model systems," *Appl. Phys. Lett.* **91**(13), 133513 (2007).
- ²⁴³J.-Y. Chen, C.-L. Hsin, C.-W. Huang, C.-H. Chiu, Y.-T. Huang, S.-J. Lin, W.-W. Wu, and L.-J. Chen, "Dynamic evolution of conducting nanofilament in resistive switching memories," *Nano Lett.* **13**(8), 3671–3677 (2013).
- ²⁴⁴I. Fina and F. Sánchez, "Seeing ferroelectric phase transitions," *Nat. Mater.* **23**(8), 1015–1016 (2024).
- ²⁴⁵K. P. Kelley, A. N. Morozovska, E. A. Eliseev, Y. Liu, S. S. Fields, S. T. Jaszewski, T. Mimura, S. Calderon, E. C. Dickey, J. F. Ihlefeld, and S. V. Kalinin, "Ferroelectricity in hafnia controlled via surface electrochemical state," *Nat. Mater.* **22**(9), 1144–1151 (2023).
- ²⁴⁶K. Humood, S. Saylan, M. Abi Jaoude, B. Mohammad, and F. Ravaux, "Impact of vacuum on the resistive switching in HfO₂-based conductive-bridge RAM with highly-doped silicon bottom electrode," *Mater. Sci. Eng.: B* **271**, 115267 (2021).
- ²⁴⁷S. M. Hus and A.-P. Li, "Spatially-resolved studies on the role of defects and boundaries in electronic behavior of 2D materials," *Prog. Surf. Sci.* **92**(3), 176–201 (2017).
- ²⁴⁸D. Mikulik, M. Ricci, G. Tutuncuoglu, F. Matteini, J. Vukajlovic, N. Vulic, E. Alarcon-Llado, and A. Fontcuberta i Morral, "Conductive-probe atomic force microscopy as a characterization tool for nanowire-based solar cells," *Nano Energy* **41**, 566–572 (2017).
- ²⁴⁹F. Giannazzo, E. Schilirò, G. Greco, and F. Roccaforte, "Conductive atomic force microscopy of semiconducting transition metal dichalcogenides and heterostructures," *Nanomaterials* **10**(4), 803 (2020).
- ²⁵⁰I. Köymen, P. Aydoğan Göktürk, C. Kocabaş, and Ş. Süzer, "Chemically addressed switching measurements in graphene electrode memristive devices using *in situ* XPS," *Faraday Discuss.* **213**, 231–244 (2019).
- ²⁵¹W. Chen, Y. Cheng, J. Ge, Z. Ma, X. Cao, S. Diao, Z. Liu, and S. Pan, "Intrinsic resistive switching in ultrathin SiO_x memristors for neuromorphic inference accelerators," *Appl. Surf. Sci.* **625**, 157191 (2023).
- ²⁵²S. Kumar, C. E. Graves, J. P. Strachan, E. M. Grafals, A. L. D. Kilcoyne, T. Tylliszczak, J. N. Weker, Y. Nishi, and R. S. Williams, "Direct observation of localized radial oxygen migration in functioning tantalum oxide memristors," *Adv. Mater.* **28**(14), 2772–2776 (2016).
- ²⁵³F. Lee, M. Tripathi, R. Sanchez Salas, S. P. Ogilvie, A. Amorim Graf, I. Jurewicz, and A. B. Dalton, "Localised strain and doping of 2D materials," *Nanoscale* **15**(16), 7227–7248 (2023).
- ²⁵⁴G. Di Martino and S. Tappertzhofen, "Optically accessible memristive devices," *Nanophotonics* **8**(10), 1579–1589 (2019).
- ²⁵⁵S. Tappertzhofen, G. Di Martino, and S. Hofmann, "Nanoparticle dynamics in oxide-based memristive devices," *Phys. Status Solidi A* **217**(6), 1900587 (2020).
- ²⁵⁶B. Salonikidou, B. Gaggio, A. Jan, M. O. Hill, G. I. Lampronti, S. Fairclough, G. D. Martino, M. Hellenbrand, and J. L. MacManus-Driscoll, "Na⁺-doped WO₃ double-layer resistive switching device for biomimetic applications," *Appl. Mater. Today* **41**, 102515 (2024).
- ²⁵⁷J. Zhang, L. Zhang, and W. Xu, "Surface plasmon polaritons: Physics and applications," *J. Phys. D: Appl. Phys.* **45**(11), 113001 (2012).
- ²⁵⁸E. Petryayeva and U. J. Krull, "Localized surface plasmon resonance: Nanostructures, bioassays and biosensing—A review," *Anal. Chim. Acta* **706**(1), 8–24 (2011).
- ²⁵⁹G. Di Martino, S. Tappertzhofen, S. Hofmann, and J. Baumberg, "Nanoscale plasmon-enhanced spectroscopy in memristive switches," *Small* **12**(10), 1334–1341 (2016).
- ²⁶⁰G. Di Martino, A. Demetriadou, W. Li, D. Kos, B. Zhu, X. Wang, B. De Nijs, H. Wang, J. MacManus-Driscoll, and J. J. Baumberg, "Real-time *in situ* optical

tracking of oxygen vacancy migration in memristors,” *Nat. Electron.* **3**(11), 687–693 (2020).

²⁶¹J. Symonowicz, “Development of brain-inspired computing,” *J. Eng. Integr.* **53**, 30–31 (2023).

²⁶²J. Symonowicz, A. Jan, H. Yan, M. Chhowalla, and G. Di Martino, “Scanning plasmon-enhanced microscopy for simultaneous optoelectrical characterization,” *ACS Nano* **18**(31), 20412–20421 (2024).

²⁶³A. Emboras, I. Goykhman, B. Desiatov, N. Mazurski, L. Stern, J. Shappir, and U. Levy, “Nanoscale plasmonic memristor with optical readout functionality,” *Nano Lett.* **13**(12), 6151–6155 (2013).

²⁶⁴A. Emboras, J. Niegemann, P. Ma, C. Haffner, A. Pedersen, M. Luisier, C. Hafner, T. Schimmel, and J. Leuthold, “Atomic scale plasmonic switch,” *Nano Lett.* **16**(1), 709–714 (2016).

²⁶⁵A. Emboras, A. Alabastri, F. Ducry, B. Cheng, Y. Salamin, P. Ma, S. Andermatt, B. Baeuerle, A. Josten, C. Hafner, M. Luisier, P. Nordlander, and J. Leuthold, “Atomic scale photodetection enabled by a memristive junction,” *ACS Nano* **12**(7), 6706–6713 (2018).

²⁶⁶C. Xu, A. J. Merryweather, S. S. Pandurangi, Z. Lun, D. S. Hall, V. S. Deshpande, N. A. Fleck, C. Schnedermann, A. Rao, and C. P. Grey, “Operando visualization of kinetically induced lithium heterogeneities in single-particle layered Ni-rich cathodes,” *Joule* **6**(11), 2535–2546 (2022).

²⁶⁷A. J. Merryweather, C. Schnedermann, Q. Jacquet, C. P. Grey, and A. Rao, “Operando optical tracking of single-particle ion dynamics in batteries,” *Nature* **594**(7864), 522–528 (2021).

²⁶⁸J. Ortega-Arroyo and P. Kukura, “Interferometric scattering microscopy (iSCAT): New frontiers in ultrafast and ultrasensitive optical microscopy,” *Phys. Chem. Chem. Phys.* **14**(45), 15625 (2012).

²⁶⁹M. Delor, H. L. Weaver, Q. Yu, and N. S. Ginsberg, “Imaging material functionality through three-dimensional nanoscale tracking of energy flow,” *Nat. Mater.* **19**(1), 56–62 (2020).

²⁷⁰R. Wang, H. Zhang, Q. Liu, F. Liu, X. Han, X. Liu, K. Li, G. Xiao, J. Albert, X. Lu, and T. Guo, “Operando monitoring of ion activities in aqueous batteries with plasmonic fiber-optic sensors,” *Nat. Commun.* **13**(1), 547 (2022).